

Improving Length Generalization in Speech Enhancement Using Relative Positional Embedding in Transformer Networks

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Abstract

Transformer-based speech enhancement models have achieved impressive performance on standard benchmarks, yet their deployment on real-world audio of varying duration remains challenging. Models trained on short fixed-length segments (typically 1–4 seconds) often exhibit severe performance degradation when applied to longer signals during inference—a problem known as the *length generalization problem*. This degradation originates primarily from the positional encoding mechanism: absolute positional encodings assign representations tied to specific position indices, creating an out-of-distribution input when the model encounters positions beyond its training range.

This thesis proposes two complementary methods to address this limitation. The first is a *Convolutional Relative Position Encoding with Logarithmic Relative Attention Bias* (CRPE + LRAB), which replaces absolute positional encoding with a combination of translation-equivariant local context from depthwise separable convolutions and a logarithmic distance bias that produces bounded attention modulation even at distances far exceeding the training range. The second is a *Sigmoid-Compressed Complex Mask* (SCCM), which extends conventional magnitude-only masks by jointly estimating magnitude and phase corrections through bounded activation functions, ensuring stable output regardless of input sequence length.

The proposed methods are evaluated using the VoiceBank-DEMAND dataset within a systematic experimental framework. All models are trained exclusively on 1-second speech segments and evaluated on inputs of 1, 5, 10, 15, and 25 seconds. A controlled ablation study with four model configurations isolates the contribution of each component. Results demonstrate that the baseline model with sinusoidal absolute positional encoding retains only 56.6% of its 1-second PESQ score at 25 seconds, falling below the noisy baseline and actively degrading the signal. The proposed complete system (CRPE + LRAB + SCCM) retains 86.6% of its 1-second PESQ at 25 seconds, representing an improvement of 30.0 percentage points in relative performance retention. The CRPE with logarithmic bias is the dominant contributor (28.3 percentage points), while the SCCM provides an additional 1.7 percentage points. These results are consistent across five complementary metrics (PESQ, ESTOI, CSIG, CBAK, COVL) and six SNR conditions ranging from −5 to 20 dB.

The findings demonstrate that the choice of positional representation has a profound impact on length generalization in Transformer-based speech enhancement, and that relative positional methods based on logarithmic distance compression provide a principled and effective solution to this problem.

Keywords: speech enhancement, Transformer, relative positional encoding, length generalization, complex mask, self-attention, deep learning

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List of Abbreviations

ALiBi	Attention with Linear Biases
CBAK	Background Intrusiveness Measure
cIRM	Complex Ideal Ratio Mask
CNN	Convolutional Neural Network
COVL	Overall Quality Measure
CRPE	Convolutional Relative Position Encoding
CSIG	Signal Distortion Measure
DEMAND	Diverse Environments Multichannel Acoustic Noise Database
DNN	Deep Neural Network
ESTOI	Extended Short-Time Objective Intelligibility
FFN	Feed-Forward Network
FFT	Fast Fourier Transform
GELU	Gaussian Error Linear Unit
GRU	Gated Recurrent Unit
IBM	Ideal Binary Mask
IRM	Ideal Ratio Mask
ISTFT	Inverse Short-Time Fourier Transform
LMMSE	Linear Minimum Mean-Square Error
LRAB	Logarithmic Relative Attention Bias
LSTM	Long Short-Term Memory
MHSA	Multi-Head Self-Attention
MMSE-LSA	Minimum Mean-Square Error Log-Spectral Amplitude
MMSE-STSA	Minimum Mean-Square Error Short-Time Spectral Amplitude
MOS	Mean Opinion Score
PE	Positional Encoding / Positional Embedding

PESQ	Perceptual Evaluation of Speech Quality
PSM	Phase-Sensitive Mask
RNN	Recurrent Neural Network
RoPE	Rotary Positional Embedding
RPR	Relative Performance Retention
SCCM	Sigmoid-Compressed Complex Mask
SI-SNR	Scale-Invariant Signal-to-Noise Ratio
SNR	Signal-to-Noise Ratio
STFT	Short-Time Fourier Transform
VAD	Voice Activity Detector

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Chapter 1. Background, Related Research, and Problem Formulation

Speech enhancement is a fundamental problem in speech processing that aims to recover a clean speech signal from an observed signal that has been degraded by additive noise, reverberation, or other distortions. Over the past five decades, speech enhancement has evolved from classical signal-processing techniques grounded in spectral estimation theory to sophisticated deep-learning architectures capable of learning complex nonlinear mappings between noisy and clean speech representations. This chapter provides a comprehensive review of the research landscape relevant to this thesis. Section 1.1 reviews traditional and neural-network-based speech enhancement methods, with particular emphasis on Transformer-based architectures. Section 1.2 examines the Transformer architecture in detail and surveys positional embedding strategies that have been proposed to encode sequential order in self-attention mechanisms. Section 1.3 reviews mask-based speech enhancement, covering the principal mask formulations used in time-frequency domain processing. Finally, Section 1.4 identifies the research gap addressed by this thesis, formulates the research questions, and states the objectives and contributions of the proposed work.

1.1 Research on Speech Enhancement

Speech enhancement encompasses a broad class of signal-processing and machine-learning techniques designed to improve the quality and intelligibility of speech signals corrupted by environmental noise. The general formulation assumes an observed noisy signal $y(t)$ that is a mixture of a clean speech signal $x(t)$ and an additive noise signal $n(t)$:

$$y(t) = x(t) + n(t) \tag{1.1}$$

where t denotes time. The objective of speech enhancement is to obtain an estimate $\hat{x}(t)$ of the clean speech signal given only the observed noisy signal $y(t)$. This section reviews two broad categories of approaches: traditional signal-processing methods and neural-network-based methods.

1.1.1 Speech Enhancement Using Traditional Signal Processing

Traditional speech enhancement methods rely on statistical models of speech and noise signals and operate primarily in the short-time Fourier transform (STFT) domain. These methods exploit the assumption that speech and noise have different spectral characteristics that can be estimated and separated using statistical estimation theory.

Spectral Subtraction

Spectral subtraction, introduced by Boll [1979], is one of the earliest and most widely cited speech enhancement techniques. The method operates on the magnitude spectrum of the noisy signal and subtracts an estimate of the noise power spectrum to obtain an estimate of the clean speech power spectrum. Applying the STFT to Equation (1.1), the noisy signal in the time-frequency domain is expressed as:

$$Y(t, f) = X(t, f) + N(t, f) \quad (1.2)$$

where $Y(t, f)$, $X(t, f)$, and $N(t, f)$ are the STFT representations of the noisy, clean, and noise signals, respectively, with t indexing the time frame and f indexing the frequency bin. Under the assumption that speech and noise are uncorrelated, the power spectrum of the noisy signal satisfies:

$$|Y(t, f)|^2 \approx |X(t, f)|^2 + |N(t, f)|^2 \quad (1.3)$$

The power spectral subtraction rule estimates the clean speech power spectrum as:

$$|\hat{X}(t, f)|^2 = |Y(t, f)|^2 - |\hat{N}(f)|^2 \quad (1.4)$$

where $|\hat{N}(f)|^2$ is an estimate of the noise power spectrum, typically obtained by averaging the power spectrum during non-speech segments detected by a voice activity detector (VAD). To prevent negative power spectral values, which are physically meaningless, half-wave rectification is applied:

$$|\hat{X}(t, f)|^2 = \max \left(|Y(t, f)|^2 - \alpha |\hat{N}(f)|^2, \beta |\hat{N}(f)|^2 \right) \quad (1.5)$$

where $\alpha \geq 1$ is an oversubtraction factor that controls the trade-off between noise suppression and speech distortion, and β is a spectral floor parameter that prevents complete spectral nulling and reduces musical noise artifacts. The enhanced signal is reconstructed by combining the estimated magnitude spectrum with the noisy phase:

$$\hat{X}(t, f) = |\hat{X}(t, f)| \cdot e^{j\angle Y(t, f)} \quad (1.6)$$

Spectral subtraction is computationally efficient, straightforward to implement, and requires no prior training. However, it suffers from several well-documented limitations. The method introduces musical noise—isolated spectral peaks that produce tonal artifacts perceptible to human listeners [Berouti et al., 1979]. It assumes noise stationarity, which limits its effectiveness in real-world environments where noise characteristics change over time. Furthermore, the use of the noisy phase in reconstruction introduces phase distortion that degrades perceptual quality, particularly at low SNR conditions.

Wiener Filtering

The Wiener filter [Wiener, 1949; Lim and Oppenheim, 1979] provides the optimal linear minimum mean-square error (LMMSE) estimate of the clean speech signal in the frequency domain. Given the noisy observation $Y(t, f)$, the Wiener filter estimate of the clean speech spectrum is:

$$\hat{X}(t, f) = H_W(t, f) \cdot Y(t, f) \quad (1.7)$$

where $H_W(t, f)$ is the Wiener filter gain, defined as:

$$H_W(t, f) = \frac{P_X(t, f)}{P_X(t, f) + P_N(t, f)} \quad (1.8)$$

Here, $P_X(t, f) = \mathbb{E}[|X(t, f)|^2]$ and $P_N(t, f) = \mathbb{E}[|N(t, f)|^2]$ denote the power spectral densities of the clean speech and noise signals, respectively. The derivation follows from minimizing the expected squared error between the estimated and true clean spectra:

$$H_W(t, f) = \arg \min_H \mathbb{E} [|X(t, f) - H \cdot Y(t, f)|^2] \quad (1.9)$$

Expressing the Wiener gain in terms of the a priori SNR, $\xi(t, f) = P_X(t, f)/P_N(t, f)$, yields the equivalent form:

$$H_W(t, f) = \frac{\xi(t, f)}{1 + \xi(t, f)} \quad (1.10)$$

The Wiener filter provides the optimal linear estimator under the assumption that speech and noise are jointly Gaussian and statistically independent. In practice, the a priori SNR must be estimated from the noisy observation, which is commonly achieved using the decision-directed approach [Ephraim and Malah, 1984]:

$$\hat{\xi}(t, f) = \alpha_{\text{dd}} \frac{|\hat{X}(t-1, f)|^2}{P_N(t, f)} + (1 - \alpha_{\text{dd}}) \max(\hat{\gamma}(t, f) - 1, 0) \quad (1.11)$$

where $\hat{\gamma}(t, f) = |Y(t, f)|^2/P_N(t, f)$ is the a posteriori SNR and α_{dd} is a smoothing constant, typically in the range $[0.95, 0.99]$. The Wiener filter produces less musical noise than spectral subtraction because the multiplicative gain varies smoothly. However, it still relies on accurate estimation of the noise power spectrum and assumes signal stationarity within each analysis frame.

Statistical-Model-Based and MMSE Approaches

Ephraim and Malah [1984] proposed an optimal estimator for the spectral amplitude of clean speech under the assumption that the STFT coefficients of both speech and noise follow complex Gaussian distributions. The MMSE short-time spectral amplitude (MMSE-STSA) estimator minimizes:

$$\hat{A}(t, f) = \mathbb{E} [|X(t, f)| | Y(t, f)] \quad (1.12)$$

The solution involves the confluent hypergeometric function and depends on the a priori and a posteriori SNRs. The MMSE-STSA estimator is given by:

$$\hat{A}(t, f) = \Gamma(1.5) \frac{\sqrt{v(t, f)}}{\hat{\gamma}(t, f)} \exp\left(-\frac{v(t, f)}{2}\right) \left[(1 + v(t, f)) I_0\left(\frac{v(t, f)}{2}\right) + v(t, f) I_1\left(\frac{v(t, f)}{2}\right) \right] \beta$$

where $v(t, f) = \frac{\xi(t, f)}{1 + \xi(t, f)} \hat{\gamma}(t, f)$, $\Gamma(\cdot)$ is the gamma function, and $I_0(\cdot)$ and $I_1(\cdot)$ are modified Bessel functions of zeroth and first order. A related estimator, the MMSE log-spectral amplitude (MMSE-LSA) estimator [Ephraim and Malah, 1985], minimizes the mean-square error of the log-spectral amplitude and yields a gain function:

$$G_{\text{LSA}}(t, f) = \frac{\xi(t, f)}{1 + \xi(t, f)} \exp\left(\frac{1}{2} \int_{v(t, f)}^{\infty} \frac{e^{-u}}{u} du\right) \quad (1.14)$$

The MMSE-LSA estimator is perceptually motivated, as the logarithmic scale aligns more closely with human auditory perception. Both estimators produce high-quality enhanced speech with reduced musical noise compared with spectral subtraction, but they remain fundamentally limited by the Gaussian assumption on the STFT coefficients, which does not accurately model the heavy-tailed distribution of speech spectral amplitudes. A comprehensive treatment of these classical methods and their theoretical foundations can be found in Loizou [2007].

Subspace Methods

An alternative class of traditional speech enhancement techniques is based on the signal subspace approach, first proposed by Ephraim and Van Trees [1995]. The fundamental insight is that clean speech occupies a lower-dimensional subspace of the observation space, while noise energy is distributed across the full space. The noisy signal covariance matrix $\mathbf{R}_y = \mathbf{R}_x + \mathbf{R}_n$ is decomposed via eigenvalue decomposition (or, equivalently, singular value decomposition), and the signal is estimated by projecting the noisy observation onto the signal subspace while suppressing the noise subspace. The enhanced signal is obtained by a linear estimator of the form:

$$\hat{\mathbf{x}} = \mathbf{H}\mathbf{y} \quad (1.15a)$$

where $\mathbf{H}$ is a filter matrix designed to minimize distortion subject to a constraint on residual noise energy. Subspace methods offer the advantage of a principled mathematical framework grounded in linear algebra and matrix perturbation theory, and they do not require explicit noise power spectrum estimation.

However, they assume local stationarity for covariance estimation and incur computational cost proportional to the cube of the frame length due to the eigenvalue decomposition, which limits their applicability in real-time systems.

Non-Negative Matrix Factorization

Non-negative matrix factorization (NMF) methods represent the magnitude spectrogram as a product of two non-negative matrices: a basis matrix $\mathbf{W} \in \mathbb{R}_{\geq 0}^{F \times K}$ containing spectral templates and an activation matrix $\mathbf{H} \in \mathbb{R}_{\geq 0}^{K \times T}$ encoding the temporal activations of each template:

$$|\mathbf{Y}| \approx \mathbf{W}\mathbf{H} = \mathbf{W}_s\mathbf{H}_s + \mathbf{W}_n\mathbf{H}_n \quad (1.15b)$$

where the subscripts s and n denote speech and noise components, respectively, and K is the total number of basis vectors. The speech and noise basis matrices can be learned from training data in a supervised manner, and at inference time, only the activations are estimated for the noisy mixture [Wilson et al., 2008]. NMF-based methods provide an interpretable decomposition of the spectrogram and can handle moderately non-stationary noise. Their primary limitation is the restricted representational capacity of the linear factorization model, which cannot capture the complex nonlinear variability of natural speech and diverse noise environments. This limitation foreshadowed the transition toward deep-learning methods that learn nonlinear representations directly from data.

Historical Perspective and Limitations of Traditional Methods

The development of traditional speech enhancement methods can be traced through several conceptual stages. The earliest approaches, exemplified by spectral subtraction [Boll, 1979] and Wiener filtering [Lim and Oppenheim, 1979], relied on direct spectral estimation with strong stationarity assumptions. The statistical-model era, initiated by Ephraim and Malah [1984, 1985], introduced principled Bayesian estimation frameworks that yielded improved perceptual quality but remained tied to parametric distributional assumptions. Subspace methods [Ephraim and Van Trees, 1995] offered a geometric perspective on signal and noise separation, while NMF methods introduced data-driven spectral templates that foreshadowed the pattern-learning paradigm of neural networks.

Time-Frequency Masking

A unifying perspective across many classical speech enhancement methods is that of time-frequency masking. Both the Wiener filter and the MMSE-STSA estimator can be interpreted as applying a time-frequency-varying gain—a mask—to the noisy spectrogram. The general masking framework defines the enhanced speech representation as:

$$\hat{X}(t, f) = M(t, f) \cdot Y(t, f) \quad (1.15)$$

where $M(t, f) \in [0, 1]$ is a real-valued mask representing the attenuation applied to each time-frequency bin. This masking perspective serves as a conceptual bridge between traditional signal-processing methods and modern deep-learning approaches, as many neural-network-based speech enhancement systems are designed to estimate an appropriate time-frequency mask rather than directly predicting the clean signal spectrum. The masking framework provides a natural decomposition of the enhancement task into estimation (predicting the mask) and reconstruction (applying the mask to the noisy signal), which has proven highly effective in both traditional and learned systems.

Despite their theoretical elegance and practical utility, traditional speech enhancement methods share fundamental limitations. They rely on statistical models that imperfectly capture the complexity of real speech and noise signals. They typically assume noise stationarity or slow variation, which is violated in many real-world acoustic environments. They operate on local spectral features without effectively exploiting long-range temporal dependencies in speech. These limitations motivated the development of data-driven approaches based on neural networks, which can learn complex, nonlinear mappings from noisy to clean speech representations directly from training data.

1.1.2 Speech Enhancement Using Neural Networks

The application of deep neural networks to speech enhancement has led to dramatic improvements in enhancement quality over the past decade. Neural-network-based methods learn the mapping between noisy and clean speech representations from large training datasets, overcoming many of the assumptions and limitations of traditional signal-processing approaches.

DNN-Based Approaches

The application of deep neural networks to speech processing marked a paradigm shift from hand-crafted statistical models to data-driven representation learning. Wang and Wang [2012] were among the first to demonstrate that a deep neural network could be trained to estimate an ideal binary mask from noisy spectral features, reframing speech enhancement as a supervised classification problem at the time-frequency level. This work established the *masking approach* as a viable DNN training target and catalyzed a wave of subsequent research. Xu et al. [2015] extended this paradigm by training DNNs to learn a direct *mapping* from noisy log-power spectral features to clean log-power spectral features, treating enhancement as a regression problem. The input to the DNN typically consists of a context window of consecutive spectral frames to incorporate limited temporal context. The DNN is trained to minimize the mean-square error between the predicted and target clean spectral features:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{TF} \sum_{t=1}^T \sum_{f=1}^F \left| \hat{X}(t, f) - X(t, f) \right|^2 \quad (1.16)$$

where T is the number of time frames, F is the number of frequency bins, and $X(t, f)$ and $\hat{X}(t, f)$ represent the target and predicted clean spectral features, respectively. An important debate in this early period concerned the choice of training target: whether the network should learn to predict the clean spectrum directly (spectral mapping) or estimate a time-frequency mask that is applied to the noisy spectrum (masking). Comparative studies [Wang et al., 2014] demonstrated that mask-based targets generally yield better generalization to unseen noise conditions, because the mask implicitly normalizes the prediction relative to the input, whereas direct mapping must learn the full spectral structure of clean speech. This finding established the mask estimation paradigm that dominates contemporary neural speech enhancement.

DNN-based methods demonstrated that learned representations could substantially outperform traditional estimators, particularly in non-stationary noise conditions where the statistical assumptions of classical methods are violated. However, frame-level DNNs have limited receptive fields and cannot model long-range temporal dependencies without excessively large context windows, which increases the number of parameters and may lead to overfitting. Furthermore, the frame-by-frame independence assumption ignores the rich temporal continuity of speech, motivating the development of architectures with explicit sequential modeling capabilities.

CNN-Based Approaches

Convolutional neural networks (CNNs) extend the capabilities of feedforward networks by applying learned convolutional filters across the time and frequency dimensions of the spectrogram. Park and Lee [2017] proposed a fully convolutional architecture that processes the noisy spectrogram end-to-end, exploiting the two-dimensional structure of time-frequency representations. Fu et al. [2018] demonstrated that CNNs with multi-scale convolutions can capture local spectro-temporal patterns that are characteristic of speech and noise at different resolutions. The convolutional architecture provides three key inductive biases: translation invariance (the same filter detects a pattern regardless of its position in the spectrogram), parameter sharing (reducing the number of learnable parameters relative to fully connected networks), and a hierarchically expanding receptive field through multiple stacked layers.

U-Net architectures [Ronneberger et al., 2015], originally developed for biomedical image segmentation, have been successfully adapted for speech enhancement by treating the spectrogram as a two-dimensional image and using skip connections between the encoder and decoder to preserve fine-grained spectral detail [Jansson et al., 2017; Choi et al., 2019]. The skip connections are particularly important for speech enhancement because they allow high-resolution spectral information to bypass the information bottleneck at the deepest layers, facilitating precise reconstruction of harmonic structure.

A significant architectural advance was the *Convolutional Recurrent Network* (CRN) proposed by Tan and Wang [2019], which combines the local feature extraction capability of CNNs with the temporal modeling capacity of recurrent layers. The CRN employs a convolutional encoder to extract frame-level feature representations, processes these features through LSTM layers to capture temporal dependencies, and reconstructs the enhanced spectrum through a convolutional decoder with skip connections. This hybrid architecture addresses the limited temporal receptive field of pure CNNs while benefiting from the efficient local processing of convolutional layers. Hu et al. [2020] extended this approach with the Deep Complex Convolution Recurrent Network (DCCRN), which applies complex-valued convolutions to jointly model the real and imaginary components of the STFT, achieving state-of-the-art results on the Interspeech 2020 Deep Noise Suppression Challenge. The DCCRN demonstrated that incorporating complex-valued operations into the convolutional framework substantially improves phase modeling without the instability associated with unconstrained complex mask estimation.

Despite their effectiveness, CNN-based architectures—including hybrid CNN-RNN designs—are inherently limited in their ability to model *global* temporal dependencies. The effective receptive field of a convolutional stack grows linearly with depth (or logarithmically with dilated convolutions), making it difficult to capture long-range correlations without very deep architectures or extremely large kernel sizes. This limitation motivates the use of attention mechanisms, which provide direct access to all positions in the sequence.

RNN/LSTM/GRU-Based Approaches

Recurrent neural networks (RNNs) and their gated variants—long short-term memory (LSTM) networks [Hochreiter and Schmidhuber, 1997] and gated recurrent units (GRUs) [Cho et al., 2014]—are natural architectures for sequential data and have been extensively applied to speech enhancement. Unlike feedforward DNNs, RNNs maintain a hidden state that is updated at each time step, enabling them to model temporal dependencies across the entire sequence. Weninger et al. [2015] and Erdogan et al. [2015] demonstrated that bidirectional LSTMs achieve significant improvements in speech enhancement quality by processing the sequence in both forward and backward directions, thereby capturing both past and future context.

The LSTM cell addresses the vanishing gradient problem of standard RNNs through its gating mechanism, which controls the flow of information through input, forget, and output gates:

$$\begin{aligned}
\mathbf{i}_t &= \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \\
\mathbf{f}_t &= \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \\
\mathbf{o}_t &= \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \\
\tilde{\mathbf{c}}_t &= \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \\
\mathbf{c}_t &= \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \\
\mathbf{h}_t &= \mathbf{o}_t \odot \tanh(\mathbf{c}_t)
\end{aligned} \tag{1.17}$$

where $\sigma(\cdot)$ is the sigmoid activation, $\odot$ denotes element-wise multiplication, and $\mathbf{i}_t$, $\mathbf{f}_t$, $\mathbf{o}_t$ are the input, forget, and output gates, respectively. Despite their ability to model temporal dependencies, RNNs suffer from sequential processing that prevents parallelization during training, and they can still struggle to capture very long-range dependencies in practice, as the effective memory of LSTM networks diminishes over extended time spans [Khandelwal et al., 2018].

Time-Domain and Frequency-Domain Processing

Neural speech enhancement methods can be categorized by their signal representation domain. Frequency-domain methods operate on the STFT representation and typically estimate a mask or directly predict the clean magnitude spectrum, using the noisy phase or an estimated phase for reconstruction. Time-domain methods operate directly on the raw waveform, bypassing the STFT and learning end-to-end from waveform to waveform. Conv-TasNet [Luo and Mesgarani, 2019], originally developed for speech separation, exemplifies the time-domain approach by using a learned encoder (a 1-D convolution) to transform the waveform into a latent representation, a separation network to estimate source masks in the latent space, and a learned decoder (a transposed 1-D convolution) to reconstruct the time-domain signals. Time-domain methods avoid the phase estimation problem and can potentially learn representations that are better adapted to the enhancement task than the STFT. However, frequency-domain methods benefit from the well-understood structure of the STFT representation and the physical interpretability of spectral features, and they remain competitive with or superior to time-domain methods in many benchmarks [Yin et al., 2020].

Complex-Valued Neural Networks

A significant limitation of methods that estimate only the magnitude mask is the loss of phase information. The phase of the noisy signal is often used directly for reconstruction, based on the assumption that phase is relatively unimportant for perceptual quality. However, research has demonstrated that phase estimation can significantly improve enhancement quality, particularly at low SNRs [Paliwal et al., 2011]. Complex-valued neural networks address this limitation by jointly estimating both the real and imaginary components of the clean STFT, or equivalently, by estimating a complex-valued mask that modifies both magnitude and phase. Williamson et al. [2016] and Choi et al. [2019] proposed deep complex U-Net architectures that process the real and imaginary parts of the spectrogram

as separate input channels and predict a complex ratio mask. This approach enables joint magnitude and phase modification, leading to improved reconstruction quality.

Mask-Based Neural Speech Enhancement

The majority of successful neural speech enhancement systems are mask-based: the neural network estimates a time-frequency mask that is applied to the noisy signal representation. This approach has several advantages. First, the mask constrains the output space, providing an inductive bias that facilitates training. Second, the mask can be interpreted as a soft selection of time-frequency bins dominated by speech, which is consistent with the auditory masking effects studied in psychoacoustics. Third, the mask-based framework decouples the estimation problem (predicting the mask) from the reconstruction problem (applying the mask), allowing the neural network to focus on learning the spectral characteristics that distinguish speech from noise. The choice of mask formulation—binary, ratio, complex, or phase-sensitive—affects both the theoretical quality ceiling and the practical difficulty of estimation, as discussed in detail in Section 1.3.

Transformer-Based Speech Enhancement

The Transformer architecture [Vaswani et al., 2017], originally developed for machine translation, has become increasingly influential in speech enhancement due to its ability to model global dependencies through self-attention and its parallelizable training procedure. Unlike RNNs, which process sequences element by element, the Transformer computes attention over the entire input sequence simultaneously, enabling it to capture long-range dependencies without the information bottleneck of recurrent hidden states.

Several Transformer-based architectures have been proposed for speech enhancement and separation. Kim et al. [2020] proposed T-GSA (Transformer with Gaussian-weighted Self-Attention), which applies a Gaussian weighting to the attention matrix to impose an inductive bias toward local temporal context while retaining the ability to model longer-range dependencies. Koizumi et al. [2020] demonstrated that Transformer encoders combined with self-adaptation mechanisms and multi-head self-attention can achieve competitive or superior performance compared with LSTM-based systems on standard speech enhancement benchmarks. The speech separation Transformer (SepFormer) [Subakan et al., 2021] employs a dual-path architecture in which short-range and long-range Transformer modules operate on overlapping chunks of the encoded signal, enabling efficient processing of long sequences while maintaining global context. The two-stage Transformer neural network (TSTNN) [Wang et al., 2021] applies Transformer layers in both the time and frequency dimensions of the spectrogram, capturing spectro-temporal dependencies at multiple scales.

The Conformer Architecture

A particularly influential development has been the Conformer architecture [Gulati et al., 2020], which addresses the local-versus-global modeling trade-off by interleaving convolutional and self-attention modules within each encoder layer. The Conformer layer applies a half-step feed-forward module, a multi-head self-attention module, a convolution module, and a second half-step feed-forward module in sequence, with residual connections around each. This design explicitly combines the global context modeling of self-attention with the local feature extraction of convolutions, achieving a balance that neither architecture can provide alone. Originally proposed for automatic speech recognition, the Conformer has been widely adopted in speech enhancement [Chen et al., 2022], and its design principle—that both local and global temporal patterns are important for speech processing—directly informs the architectural choices made in this thesis.

The Local-Global Dependency Trade-Off

The evolution from DNNs through CNNs, RNNs, and Transformers to hybrid architectures such as the Conformer can be understood through the lens of the *local-global dependency trade-off*. DNNs capture only the dependencies within a fixed context window. CNNs efficiently extract local spectro-temporal patterns but require deep stacking to expand the receptive field to global scale. RNNs and LSTMs model sequential dependencies through hidden state propagation but suffer from the information bottleneck of a fixed-size hidden vector and the practical limitation of vanishing gradients over long time spans. Pure Transformer encoders provide direct global access through self-attention but lack the inductive bias toward local structure that convolutions provide, and they incur quadratic computational cost. The most successful recent architectures combine these paradigms: the CRN [Tan and Wang, 2019] combines convolution with recurrence, and the Conformer [Gulati et al., 2020] combines convolution with self-attention.

The Transformer architecture offers several advantages for speech enhancement. Its self-attention mechanism provides a direct mechanism for modeling long-range temporal dependencies, which is important for capturing the temporal structure of speech phonemes, prosody, and speaker characteristics. The multi-head attention mechanism enables the model to simultaneously attend to different aspects of the input signal at different representation subspaces. The parallelizable architecture enables efficient training on modern GPU hardware.

However, Transformer-based speech enhancement models face several important challenges. The self-attention mechanism has $O(L^2)$ computational and memory complexity with respect to the sequence length L , which limits the maximum sequence length that can be processed, particularly during training. Positional information must be explicitly injected into the model because the self-attention operation is permutation-equivariant and has no inherent notion of sequential order. Most critically for this thesis, conventional positional encoding methods—particularly absolute positional encodings—are designed for a fixed

maximum sequence length and may not generalize well when the model is applied to sequences substantially longer than those observed during training. This *length generalization problem* is the central challenge addressed by this thesis and is discussed in detail in the following sections.

Critical Synthesis: The Unaddressed Problem of Length Generalization

Despite the substantial progress in neural speech enhancement architectures, a critical observation emerges from the literature reviewed above: virtually all proposed systems are evaluated exclusively on signals whose duration matches the training segment length. Standard benchmarks such as VoiceBank-DEMAND [Valentini-Botinhao et al., 2016] consist of utterances that are typically 1–4 seconds in duration, and published results report performance only at these in-distribution lengths. The implicit assumption is that models will be applied to signals of the same duration as those seen during training, or that longer signals will be chunked into short overlapping segments and processed independently.

This assumption is problematic for two reasons. First, independent chunk-based processing introduces boundary artifacts at the transitions between segments and fails to exploit contextual information that spans multiple chunks. Second, and more fundamentally, the Transformer’s positional encoding mechanism creates an explicit dependence on the sequence length observed during training: absolute positional encodings assign representations to specific position indices, and the model has no mechanism to produce meaningful outputs at positions it has never encountered. No published work in the speech enhancement literature has systematically investigated this length generalization failure or proposed methods specifically designed to address it. This gap is the primary motivation for the research presented in this thesis.

1.2 Transformer Architecture and Positional Embedding

The Transformer architecture [Vaswani et al., 2017] has fundamentally transformed sequence modeling across natural language processing, speech processing, and computer vision. This section provides a detailed description of the Transformer architecture and reviews the positional embedding methods that have been proposed to encode sequential order in self-attention-based models.

1.2.1 The Transformer Architecture

Scaled Dot-Product Attention

The core computational mechanism of the Transformer is the scaled dot-product attention. Given an input sequence represented as a matrix $\mathbf{Z} \in \mathbb{R}^{L \times d}$, where L is the sequence length and d is the feature dimension, the input is linearly projected into three distinct representations: queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$:

$$\mathbf{Q} = \mathbf{Z}\mathbf{W}^Q, \quad \mathbf{K} = \mathbf{Z}\mathbf{W}^K, \quad \mathbf{V} = \mathbf{Z}\mathbf{W}^V \quad (1.18)$$

where $\mathbf{W}^Q, \mathbf{W}^K \in \mathbb{R}^{d \times d_k}$ and $\mathbf{W}^V \in \mathbb{R}^{d \times d_v}$ are learned projection matrices, and d_k and d_v are the dimensions of the key and value spaces, respectively. The attention output is computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (1.19)$$

The scaling factor $\sqrt{d_k}$ prevents the dot products from growing excessively large as the dimensionality increases, which would push the softmax function into regions of extremely small gradients [Vaswani et al., 2017]. The attention matrix $\mathbf{A} = \text{softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d_k}) \in \mathbb{R}^{L \times L}$ represents the pairwise attention weights, where each element a_{ij} indicates the degree to which position i attends to position j . A critical observation is that this attention computation is *permutation-equivariant*: permuting the input positions and correspondingly permuting the queries, keys, and values produces the same output with the same permutation applied. Consequently, the attention mechanism alone has no notion of sequential order, which motivates the need for positional information.

Multi-Head Attention

Rather than performing a single attention computation, the Transformer employs multi-head attention, which projects the input into h parallel attention heads, each operating in a lower-dimensional subspace:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O \quad (1.20)$$

where each head is computed as:

$$\text{head}_i = \text{Attention}(\mathbf{Z}\mathbf{W}_i^Q, \mathbf{Z}\mathbf{W}_i^K, \mathbf{Z}\mathbf{W}_i^V) \quad (1.21)$$

with $\mathbf{W}_i^Q, \mathbf{W}_i^K \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}_i^V \in \mathbb{R}^{d \times d_v}$, and $\mathbf{W}^O \in \mathbb{R}^{hd_v \times d}$. Each attention head can learn to attend to different aspects of the input, such as local patterns in one head and long-range dependencies in another. This provides the model with a richer representational capacity compared with single-head attention.

Position-Wise Feed-Forward Network

Each Transformer layer includes a position-wise feed-forward network (FFN) that applies two linear transformations with a nonlinear activation:

$$\text{FFN}(\mathbf{z}) = \mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2 \quad (1.22)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d \times d_{\text{ff}}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$, and d_{ff} is the inner dimension, typically set to $4d$. The FFN is applied identically and independently to each position, providing element-wise nonlinear transformation that complements the sequence-wide mixing performed by the attention mechanism.

Residual Connections and Layer Normalization

Each sub-layer (multi-head attention and FFN) is wrapped with a residual connection [He et al., 2016] followed by layer normalization [Ba et al., 2016]:

$$\mathbf{z}' = \text{LayerNorm}(\mathbf{z} + \text{Sublayer}(\mathbf{z})) \quad (1.23)$$

Residual connections facilitate gradient flow through deep networks by providing a direct path for the gradient during backpropagation. Layer normalization normalizes the activations across the feature dimension for each position independently, stabilizing training and improving convergence. The combination of residual connections and layer normalization enables training of very deep Transformer models with dozens of layers.

Encoder Architecture

A Transformer encoder consists of a stack of N identical layers, each containing a multi-head self-attention sub-layer followed by a position-wise FFN sub-layer, both with residual connections and layer normalization. The encoder processes an input sequence $\mathbf{Z} \in \mathbb{R}^{L \times d}$ and produces a contextualized representation of the same dimensionality. For speech enhancement, the encoder is the primary architectural component, as the task does not require autoregressive generation: the model receives the entire noisy signal representation and produces an enhanced representation or mask in a single forward pass.

1.2.2 Positional Embedding Methods

Because the self-attention mechanism is permutation-equivariant, the Transformer requires an explicit mechanism to encode positional information. Without such a mechanism, the model cannot distinguish between different orderings of the same set of input elements, which is clearly unacceptable for sequential data such as speech. This subsection reviews the principal positional encoding methods and analyzes their suitability for speech enhancement, particularly with respect to generalization to sequence lengths not observed during training.

Sinusoidal Positional Encoding

The original Transformer [Vaswani et al., 2017] proposed sinusoidal positional encodings, where the position of each element in the sequence is encoded using sine and cosine functions of different frequencies:

$$\text{PE}(p, 2i) = \sin\left(\frac{p}{10000^{2i/d}}\right), \quad \text{PE}(p, 2i + 1) = \cos\left(\frac{p}{10000^{2i/d}}\right) \quad (1.24)$$

where p is the position index and i is the dimension index. The positional encoding is added to the input embedding:

$$\mathbf{z}_p = \mathbf{e}_p + \text{PE}(p) \quad (1.25)$$

where $\mathbf{e}_p$ is the input embedding at position p . The sinusoidal encoding has the property that the relative position between any two positions p and $p + k$ can be expressed as a linear function of $\text{PE}(p)$, since $\text{PE}(p + k)$ is a linear transformation of $\text{PE}(p)$ for any fixed offset k . While this property theoretically allows the model to attend to relative positions, in practice, the model must learn to extract this information through the attention mechanism's query-key dot products, which is an indirect and potentially unreliable process [Shaw et al., 2018].

Analysis of extrapolation failure. To understand why sinusoidal positional encodings fail to generalize to unseen lengths, consider the dot product between positional encoding vectors at positions p and q . For a single frequency component i , the contribution to the dot product is:

$$\text{PE}(p, 2i) \cdot \text{PE}(q, 2i) + \text{PE}(p, 2i+1) \cdot \text{PE}(q, 2i+1) = \cos\left(\frac{p - q}{10000^{2i/d}}\right) \quad (1.25a)$$

Summing across all $d/2$ frequency components, the full dot product $\langle \text{PE}(p), \text{PE}(q) \rangle$ depends only on the offset $p - q$. However, this mathematical property does not ensure that the *learned* attention patterns generalize. During training on sequences of length L_{train} , the attention weights $a_{ij} \propto \exp(\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k})$ are shaped by the joint distribution of content embeddings $\mathbf{e}_p$ and positional encodings $\text{PE}(p)$ for $p \in \{0, \dots, L_{\text{train}} - 1\}$. At inference on a longer sequence, queries at positions $p \geq L_{\text{train}}$ produce activations $\mathbf{q}_p = (\mathbf{e}_p + \text{PE}(p))\mathbf{W}^Q$ that lie outside the distribution the attention parameters were optimized for. The resulting attention weights become unpredictable, as the trained projection matrices $\mathbf{W}^Q$ and $\mathbf{W}^K$ have not learned to produce meaningful query-key interactions at these unseen positional offsets. Press et al. [2022] empirically confirmed that Transformers with sinusoidal positional encodings exhibit sharply degraded perplexity when evaluated on sequences even modestly longer than the training length.

Learned Positional Embeddings

An alternative approach replaces the fixed sinusoidal functions with learned positional embedding vectors:

$$\mathbf{z}_p = \mathbf{e}_p + \mathbf{p}_p \quad (1.26)$$

where $\mathbf{p}_p \in \mathbb{R}^d$ is a learned embedding vector for position p , stored in a lookup table of size $L_{\text{max}} \times d$. This approach allows the model to learn position-specific representations tailored to the training data. However, learned positional embeddings are inherently constrained to the maximum sequence length L_{max} observed during training and cannot represent positions beyond this length. This is a fundamental limitation for applications such as speech enhancement, where the model may need to process sequences of arbitrary length during inference.

Relative Positional Encoding

Shaw et al. [2018] proposed relative positional encoding, which modifies the attention mechanism to directly incorporate information about the relative distance between positions rather than adding absolute positional information to the input embeddings. The attention logit between positions i and j is modified to include a learned relative positional bias:

$$e_{ij} = \frac{\mathbf{q}_i^\top \mathbf{k}_j + \mathbf{q}_i^\top \mathbf{a}_{ij}^K + b_{ij}}{\sqrt{d_k}} \quad (1.27)$$

where $\mathbf{a}_{ij}^K$ is a learned key-side relative position embedding that depends only on the offset $j - i$, and b_{ij} is a scalar bias. The relative position embeddings are typically clipped to a maximum distance $k_{\max}$:

$$\mathbf{a}_{ij}^K = \mathbf{r}_{\text{clip}(j-i, -k_{\max}, k_{\max})} \quad (1.28)$$

This approach is attractive because the model explicitly attends based on relative distance, which is more linguistically and acoustically meaningful than absolute position. Furthermore, it provides a mechanism for generalization to unseen sequence lengths, since the relative distance between any two positions can be represented using the learned embeddings as long as the distance falls within the clipped range. However, the clipping operation introduces an information boundary beyond which all distant positions appear identical to the model.

Transformer-XL Relative Positional Encoding

Dai et al. [2019] proposed an influential variant of relative positional encoding within the Transformer-XL architecture, designed for language modeling over long documents. The Transformer-XL decomposes the standard attention score into four terms:

$$e_{ij} = \underbrace{\mathbf{q}_i^\top \mathbf{W}_E^\top \mathbf{W}_E \mathbf{k}_j}_{(a) \text{ content-content}} + \underbrace{\mathbf{q}_i^\top \mathbf{W}_E^\top \mathbf{W}_R \mathbf{r}_{i-j}}_{(b) \text{ content-position}} + \underbrace{\mathbf{u}^\top \mathbf{W}_E \mathbf{k}_j}_{(c) \text{ global content bias}} + \underbrace{\mathbf{v}^\top \mathbf{W}_R \mathbf{r}_{i-j}}_{(d) \text{ global position bias}} \quad (1.28a)$$

where $\mathbf{r}_{i-j}$ is a sinusoidal encoding of the relative distance $i - j$, $\mathbf{u}$ and $\mathbf{v}$ are learnable global bias vectors shared across all queries, and $\mathbf{W}_E$ and $\mathbf{W}_R$ are content and position projection matrices, respectively. This decomposition makes explicit the distinct roles of content-based and position-based attention, and uses relative rather than absolute position indices. The Transformer-XL architecture also introduces a segment-level recurrence mechanism that extends the effective context length across segment boundaries, although this recurrence introduces sequential dependency between segments that complicates parallelization.

Rotary Positional Embedding (RoPE)

Rotary positional embedding (RoPE) [Su et al., 2024] encodes positional information by rotating the query and key vectors in the complex plane. For a d -dimensional representation, consecutive pairs of dimensions $(2i, 2i + 1)$ are treated as two-dimensional subspaces, and a position-dependent rotation is applied to each pair. Formally, for a vector $\mathbf{x} \in \mathbb{R}^d$ at position p , the rotary embedding applies:

$$\text{RoPE}(\mathbf{x}, p) = \begin{pmatrix} x_0 \cos p\theta_0 - x_1 \sin p\theta_0 \\ x_0 \sin p\theta_0 + x_1 \cos p\theta_0 \\ \vdots \\ x_{d-2} \cos p\theta_{d/2-1} - x_{d-1} \sin p\theta_{d/2-1} \\ x_{d-2} \sin p\theta_{d/2-1} + x_{d-1} \cos p\theta_{d/2-1} \end{pmatrix} \quad (1.29)$$

where $\theta_i = 10000^{-2i/d}$ are the rotation frequencies, which decay geometrically across dimensions. This can be expressed compactly as a block-diagonal rotation matrix $\mathbf{R}_p = \text{diag}(\mathbf{R}_p^{(0)}, \dots, \mathbf{R}_p^{(d/2-1)})$ where each 2×2 block is:

$$\mathbf{R}_p^{(i)} = \begin{pmatrix} \cos p\theta_i & -\sin p\theta_i \\ \sin p\theta_i & \cos p\theta_i \end{pmatrix} \quad (1.29a)$$

The key property of RoPE is that the dot product between rotated query and key vectors at positions p and q depends only on the relative distance $p - q$:

$$\langle \mathbf{R}_p \mathbf{q}, \mathbf{R}_q \mathbf{k} \rangle = \mathbf{q}^\top \mathbf{R}_p^\top \mathbf{R}_q \mathbf{k} = \mathbf{q}^\top \mathbf{R}_{q-p} \mathbf{k} = g(\mathbf{q}, \mathbf{k}, p - q) \quad (1.29b)$$

since the product of rotation matrices yields a rotation by the difference of angles. This inherent relative-position dependence, combined with the smooth rotation-based encoding, gives RoPE strong theoretical motivation for length generalization. The geometric decay of rotation frequencies allows different dimensions to encode positional information at different scales: high-frequency dimensions capture fine-grained local position, while low-frequency dimensions encode coarse global position. RoPE has demonstrated strong performance in large language models [Touvron et al., 2023] and has been extended with various frequency-scaling strategies (NTK-aware interpolation, YaRN) to improve extrapolation to lengths far beyond the training distribution. However, its application to speech enhancement remains largely unexplored, and the rotation-based encoding introduces a subtle dependence on the absolute magnitudes of query and key vectors that may interact with the spectral dynamics of speech signals in ways that have not been characterized.

ALiBi (Attention with Linear Biases)

Press et al. [2022] proposed ALiBi, which adds a static, non-learned linear bias to the attention scores based on the distance between query and key positions:

$$a_{ij} = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_k}} - m \cdot |i - j| \quad (1.30)$$

where m is a head-specific slope that is fixed (not learned) and set according to a geometric sequence across attention heads. ALiBi does not add positional information to the input embeddings; instead, it imposes a recency bias directly on the attention weights by penalizing distant positions. This approach is simple, computationally cheap, and has demonstrated improved extrapolation to longer sequences in language modeling tasks. However, the linear decay may be overly restrictive for speech processing, where both local and long-range dependencies are important and the relevant attention patterns may not follow a monotonically decaying profile.

Comparative Analysis

Table 1.1 summarizes the key properties of the positional encoding methods reviewed above with respect to their suitability for speech enhancement with length generalization.

Table 1.1: Comparison of positional encoding methods for speech enhancement.

Method	Position Type	Length Generalization	Complexity Overhead	Relative Dist.
Sinusoidal	Absolute	Limited	None	Implicit
Learned	Absolute	None	None	None
Shaw et al.	Relative	Moderate (clipped)	Low	Explicit
Transformer-XL	Relative	Moderate	Low	Explicit
RoPE	Relative	Strong	Minimal	Explicit
ALiBi	Relative (bias)	Strong	Minimal	Explicit (linear)

Absolute positional encodings—both sinusoidal and learned—are fundamentally limited in their ability to generalize to unseen sequence lengths. Sinusoidal encodings produce positional representations at unseen positions, but the model has never observed these representations during training, and empirical evidence suggests that extrapolation performance degrades significantly [Press et al., 2022]. Learned embeddings cannot represent positions beyond the maximum training length at all.

Relative positional methods offer a more promising path toward length generalization because they encode the *relationship* between positions rather than absolute position identity. However, the different relative methods exhibit distinct trade-offs that merit careful consideration for the speech enhancement domain. Shaw et al.’s method provides explicit learned relative embeddings but

clips distances beyond $k_{\max}$, creating an information boundary that limits generalization to distances beyond the clipping range. The Transformer-XL decomposition [Dai et al., 2019] offers a principled factorization of content and position attention but introduces segment-level recurrence that complicates deployment. RoPE provides an elegant mathematical framework that naturally encodes relative distance through rotation in the complex plane, but it has been developed primarily for language modeling and its interaction with continuous spectral features has not been studied. ALiBi [Press et al., 2022] offers simplicity and demonstrated extrapolation capability, but its strictly linear distance penalty imposes a monotonic recency bias that may be too restrictive for speech signals, where relevant context can appear at variable distances (e.g., a repeated phoneme or a noise segment that informs enhancement of a later region).

A key observation from this comparative analysis is that no existing positional encoding method has been specifically designed for or evaluated in the context of speech enhancement length generalization. The methods reviewed above were developed primarily for natural language processing tasks, where the statistics of token sequences differ substantially from those of spectral frame sequences. Speech enhancement presents unique requirements: the importance of local temporal context (neighboring frames share spectral continuity), the translation-equivariant nature of the enhancement task (the same phoneme requires the same processing regardless of its absolute position in the utterance), and the need to process signals of widely varying duration without retraining. These requirements motivate the development of a positional encoding mechanism specifically tailored to the speech enhancement setting, as proposed in Chapter 2.

1.3 Masks for Speech Enhancement

The mask-based approach is the dominant paradigm in both traditional and neural-network-based speech enhancement. In this framework, the enhancement system estimates a time-frequency mask that is applied to the noisy signal representation to reconstruct the clean speech. This section reviews the principal mask formulations, their mathematical definitions, and their properties with respect to speech enhancement quality, noise suppression, and generalization.

1.3.1 Ideal Binary Mask (IBM)

The ideal binary mask [Wang, 2005] assigns each time-frequency bin a value of 1 if the local SNR exceeds a threshold θ , and 0 otherwise:

$$M_{\text{IBM}}(t, f) = \begin{cases} 1 & \text{if } \text{SNR}(t, f) > \theta \\ 0 & \text{otherwise} \end{cases} \quad (1.31)$$

where the local SNR is defined as $\text{SNR}(t, f) = 10 \log_{10} (|X(t, f)|^2 / |N(t, f)|^2)$. The IBM is motivated by the auditory masking phenomenon observed in psychoacoustics, where speech components are perceptible in a given time-frequency region only

if their energy exceeds the masking threshold of the noise. While the IBM provides effective noise suppression and can improve speech intelligibility, the binary nature of the mask introduces harsh transitions between retained and suppressed time-frequency bins, resulting in distortion of the speech signal, particularly at the boundaries between speech-dominated and noise-dominated regions.

1.3.2 Ideal Ratio Mask (IRM)

The ideal ratio mask [Srinivasan et al., 2006; Narayanan and Wang, 2013] provides a soft, continuous-valued mask based on the ratio of speech energy to total energy in each time-frequency bin:

$$M_{\text{IRM}}(t, f) = \left(\frac{|X(t, f)|^2}{|X(t, f)|^2 + |N(t, f)|^2} \right)^\beta \quad (1.32)$$

where β is a tunable exponent, commonly set to 0.5 (for the amplitude-domain ratio) or 1 (for the power-domain ratio). With $\beta = 1$, the IRM is equivalent to the Wiener filter gain. The IRM produces smoother enhanced spectra than the IBM and preserves partial speech energy in noise-dominated regions, leading to less speech distortion and fewer artifacts. The mask values are bounded in $[0, 1]$, which provides a natural constraint for neural network estimation using a sigmoid output activation.

1.3.3 Complex Masks

All of the preceding masks are real-valued and modify only the magnitude of the noisy spectrum, typically reusing the noisy phase for reconstruction. Complex masks [Williamson et al., 2016] extend the masking framework to jointly modify both magnitude and phase by estimating a complex-valued mask:

$$M_c(t, f) = M_r(t, f) + jM_i(t, f) \quad (1.33)$$

where M_r and M_i are the real and imaginary components of the mask. The enhanced complex spectrum is obtained by complex multiplication:

$$\hat{X}(t, f) = M_c(t, f) \cdot Y(t, f) = (M_r + jM_i)(Y_r + jY_i) \quad (1.34)$$

which expands to:

$$\hat{X}_r = M_r Y_r - M_i Y_i, \quad \hat{X}_i = M_r Y_i + M_i Y_r \quad (1.35)$$

Complex masks provide the theoretical capacity to perfectly reconstruct the clean complex spectrum from the noisy observation, including phase correction. The complex ideal ratio mask (cIRM) is defined as:

$$M_{\text{cIRM}}(t, f) = \frac{Y_r X_r + Y_i X_i}{|Y|^2} + j \frac{Y_r X_i - Y_i X_r}{|Y|^2} \quad (1.36)$$

which satisfies $M_{\text{cIRM}} \cdot Y = X$ exactly. Training neural networks to estimate complex masks is more challenging than estimating real-valued masks because the output is unbounded, requiring careful loss design and output activation strategies.

To address the unboundedness of the cIRM, Tan and Wang [2019] proposed *compressing* the complex mask components using a hyperbolic tangent function:

$$\hat{M}_r = K \cdot \tanh(m_r), \quad \hat{M}_i = K \cdot \tanh(m_i) \quad (1.36a)$$

where m_r and m_i are the raw network outputs and K is a compression factor (typically $K = 10$) that determines the maximum mask magnitude. This compressed complex ratio mask (CCRM) restricts the output range to $[-K, K]$ for each component, preventing the extreme values that can destabilize training while retaining sufficient dynamic range for speech restoration. The CCRM represents a practical compromise between the theoretical optimality of the unconstrained cIRM and the training stability of bounded real-valued masks.

Theoretical Capacity of Mask Formulations

The different mask formulations can be compared in terms of their theoretical reconstruction capacity. Let $\mathcal{M}$ denote the set of achievable masks for a given formulation. For the IBM, $\mathcal{M}_{\text{IBM}} = \{0, 1\}^{T \times F}$, which can only select or discard entire time-frequency bins and introduces an irreducible *quantization error* proportional to the energy of partially speech-dominated bins. For the IRM, $\mathcal{M}_{\text{IRM}} = [0, 1]^{T \times F}$, which allows continuous attenuation but cannot amplify speech energy that has been masked by noise, nor correct phase errors. For the cIRM, $\mathcal{M}_{\text{cIRM}} = \mathbb{C}^{T \times F}$, which can in principle achieve perfect reconstruction $\hat{X} = X$ for any $Y \neq 0$. However, this theoretical capacity comes at the cost of practical difficulty: as $|Y(t, f)| \rightarrow 0$ in noise-dominated regions, the ideal complex mask $M_{\text{cIRM}} = X/Y$ diverges, producing extreme target values that destabilize gradient-based optimization.

Training Stability with Unbounded Mask Targets

The practical challenge of estimating complex masks deserves particular attention because it directly motivates the mask formulation proposed in this thesis. When the cIRM is used as the training target, the network must predict values that can be arbitrarily large in magnitude, particularly in time-frequency bins where the noisy signal energy is small relative to the clean signal energy. In such bins, the mask $M = X/Y$ can exceed values of 10 or more, creating extreme outliers in the loss landscape that produce large, destabilizing gradients. Several strategies have been proposed to mitigate this issue: target clipping [Williamson et al., 2016], which discards information beyond a threshold; compression functions such as $\tanh$ [Tan and Wang, 2019]; and alternative loss formulations

that operate in the magnitude domain rather than on the mask values directly. Each strategy involves a trade-off between reconstruction fidelity (the ability to achieve the theoretical optimum) and training stability (the ability to converge reliably). The mask proposed in Section 2.4 introduces a novel approach to this trade-off that decomposes the complex mask into bounded magnitude and phase components, achieving stability without sacrificing the ability to jointly modify both magnitude and phase.

1.3.4 Phase-Sensitive Mask (PSM)

The phase-sensitive mask [Erdogan et al., 2015] provides a compromise between the simplicity of real-valued masks and the phase-awareness of complex masks:

$$M_{\text{PSM}}(t, f) = \frac{|X(t, f)|}{|Y(t, f)|} \cos(\theta_x(t, f) - \theta_y(t, f)) \quad (1.37)$$

where θ_x and θ_y are the phases of the clean and noisy spectra, respectively. The PSM is real-valued and is applied to the magnitude of the noisy signal, but it accounts for the phase difference between clean and noisy signals. This allows the mask to partially compensate for phase distortion while remaining within a scalar masking framework.

1.3.5 Mask Application and Speech Reconstruction

Regardless of the specific mask formulation, the general reconstruction procedure follows a consistent framework. Given the estimated mask $\hat{M}(t, f)$ and the noisy STFT representation $Y(t, f)$, the enhanced speech representation is obtained as:

$$\hat{X}(t, f) = \hat{M}(t, f) \odot Y(t, f) \quad (1.38)$$

where $\odot$ denotes element-wise multiplication. The enhanced time-domain signal is then reconstructed using the inverse STFT (ISTFT). The choice of mask formulation affects the quality of reconstruction through several factors: the accuracy of magnitude estimation, the degree of phase correction, the smoothness of the estimated mask, and the difficulty of the estimation task for the neural network. For the purpose of length generalization, an important consideration is whether the mask formulation introduces any implicit dependence on sequence length—for instance, through normalization operations that aggregate across the entire time dimension.

1.4 Problem Selection and Research Objectives

The literature review presented in Sections 1.1 through 1.3 reveals a significant and largely unaddressed research challenge at the intersection of Transformer-based speech enhancement and positional encoding design. While Transformer architectures have demonstrated impressive performance on standard speech enhancement benchmarks, their deployment in practical applications is

complicated by the *length generalization problem*: models trained on short audio segments often exhibit degraded performance when applied to longer signals during inference.

1.4.1 Identification of the Research Gap

The literature surveyed in the preceding sections reveals three converging lines of research—Transformer-based speech enhancement, relative positional encoding, and complex mask design—that have been pursued largely in isolation. To date, no published work has systematically investigated how these three components interact to determine the length generalization capability of a speech enhancement system. This subsection identifies the specific gaps in the existing literature that motivate the present thesis.

Gap 1: Length Generalization Is Unstudied in Speech Enhancement

In practical speech enhancement applications, models are typically trained on fixed-length audio segments of short duration (e.g., 1–4 seconds) due to the $O(L^2)$ computational and memory complexity of self-attention. However, real-world speech signals can be much longer—ranging from a few seconds in voice commands to several minutes in conversations and lectures. The standard approach of processing long signals by splitting them into short overlapping windows and independently enhancing each window introduces boundary artifacts, discontinuities, and a failure to exploit long-range contextual information that spans multiple windows. Despite the practical importance of this problem, existing speech enhancement studies evaluate models exclusively on the same duration used during training. The VoiceBank-DEMAND benchmark [Valentini-Botinhao et al., 2016], which is the dominant evaluation protocol, uses fixed-length utterances averaging approximately 2.5 seconds, providing no information about how a model would perform on inputs $5\times$ or $25\times$ longer. This stands in sharp contrast to the natural language processing literature, where length generalization has been a central concern since the work of Press et al. [2022] on ALiBi and Kazemnejad et al. [2024] on the impact of positional encoding choice.

Gap 2: Positional Encoding Choice for Audio Transformers Is Underexplored

The fundamental cause of the length generalization problem in Transformer-based models lies in the positional encoding mechanism. Absolute positional encodings—whether sinusoidal or learned—provide position representations that are inherently tied to absolute indices. A model trained exclusively with sequences of length L_{train} has never observed the positional representations corresponding to positions beyond L_{train} , and there is no guarantee that the learned attention patterns will extend meaningfully to these unseen positions. The rapid development of relative positional methods in NLP—Shaw et al. [2018], Dai et al. [2019], Su et al. [2024], Press et al. [2022]—has produced several candidates that could address this problem, yet their application to speech enhancement has been minimal. Kim et al. [2020] adopted sinusoidal absolute PE without modification; Koizumi et al. [2020] used learned absolute embeddings;

and even Conformer-based speech enhancement systems [Gulati et al., 2020] rely on absolute PE augmented with local convolution. Relative positional encodings offer a theoretically grounded solution by encoding the relationship between positions rather than absolute position identity, but the effectiveness of different relative positional methods for speech enhancement length generalization has not been systematically investigated.

Gap 3: Mask Formulation and Length Robustness Have Not Been Jointly Studied

The mask formulation determines how the network's output is converted into the enhanced speech signal, yet the interaction between mask design and length generalization has received no attention. The standard IRM, being bounded in $[0,1]$, is inherently stable but discards phase information and limits reconstruction quality. Complex-valued masks [Williamson et al., 2016] can achieve theoretically perfect reconstruction but introduce unbounded targets that may amplify errors at positions far from the training distribution. Whether the choice of mask formulation affects a model's robustness to length mismatch between training and inference is an open question that this thesis directly addresses through controlled ablation experiments.

1.4.2 Research Problem Statement

The central research problem addressed by this thesis is:

How can a Transformer-based speech enhancement model trained on short-duration speech segments maintain or improve its enhancement performance when processing significantly longer speech signals during inference?

This problem is addressed through three interconnected research components:

1. **A Transformer-based speech enhancement architecture** specifically designed for mask-based enhancement, using the Transformer encoder as the primary sequence modeling component.
2. **A relative positional embedding method** designed to improve sequence-length generalization by encoding relative relationships between positions rather than absolute position indices.
3. **A speech enhancement mask formulation** designed to provide effective reconstruction of clean speech from noisy input while being robust to variation in input sequence length.

1.4.3 Research Questions

To guide the investigation, this thesis formulates the following specific research questions:

1. **RQ1:** Does relative positional embedding improve length generalization in Transformer-based speech enhancement compared with conventional absolute positional encoding?

2. **RQ2:** How does the proposed positional embedding affect speech enhancement performance on sequences substantially longer than those used during training (e.g., 5 $\times$, 10 $\times$, 15 $\times$, and 25 $\times$ the training length)?
3. **RQ3:** Does the proposed mask formulation improve speech enhancement quality compared with standard mask formulations such as the ideal ratio mask?
4. **RQ4:** How does the combination of the proposed Transformer architecture, relative positional embedding, and mask formulation affect overall enhancement performance?
5. **RQ5:** To what extent can a model trained exclusively on 1-second speech segments generalize to 5-, 10-, 15-, and 25-second inputs without significant performance degradation?
6. **RQ6:** What is the individual contribution of each proposed component (positional embedding and mask design) to the observed performance improvements, as determined by controlled ablation experiments?

1.4.4 Research Approach and Evaluation Strategy

This thesis adopts a *quantitative empirical* research methodology, following the controlled experimental paradigm established in the speech enhancement literature [Loizou, 2007]. The investigation proceeds through a sequence of methodological steps designed to isolate the contribution of each proposed component to the overall system performance and, critically, to its length generalization capability.

Methodological Framework

The experimental framework is structured around four models that instantiate all pairwise combinations of two binary factors: (i) positional encoding type (absolute sinusoidal versus the proposed relative method) and (ii) mask formulation (standard IRM versus the proposed SCCM). This 2 $\times$ 2 factorial design enables a principled ablation study [Meyes et al., 2019] that quantifies the individual and combined effects of each component. All four models share an identical Transformer encoder backbone with the same hyperparameters ($d_{\text{model}} = 256$, $n_h = 8$, $d_{\text{ff}} = 1024$, $L = 6$ layers), and are trained under strictly identical conditions: the same training data, random seeds, loss function, optimizer, and learning-rate schedule. This design ensures that any observed performance differences can be attributed to the specific component being varied, rather than to confounding experimental factors.

Training and Evaluation Protocol

All models are trained exclusively on approximately 1-second noisy/clean speech pairs drawn from the VoiceBank-DEMAND dataset [Valentini-Botinhao et al., 2016]. The critical aspect of the evaluation protocol is the *mismatched-length testing regime*: after training, each model is evaluated on inputs of 1, 5, 10, 15, and 25 seconds in duration, representing length ratios of 1 $\times$ to 25 $\times$ relative to the training segment length. This protocol directly tests the central hypothesis of the

thesis by exposing models to sequence lengths they have never encountered during training. The 1-second evaluation serves as a within-distribution baseline, while the longer evaluations quantify the rate and severity of performance degradation as the length mismatch increases.

Evaluation Metrics

Speech enhancement quality is assessed using five complementary metrics that capture different perceptual dimensions:

1. **PESQ** (Perceptual Evaluation of Speech Quality) [Rix et al., 2001]: an ITU-T standard (P.862) that models the perceptual quality of narrowband and wideband speech, producing scores on a scale from -0.5 to 4.5 .
2. **ESTOI** (Extended Short-Time Objective Intelligibility) [Jensen and Taal, 2016]: measures the correlation between the temporal envelopes of clean and enhanced speech in one-third octave bands, producing values in $[0, 1]$.
3. **CSIG** (Signal Distortion Measure) [Hu and Loizou, 2008]: predicts the mean opinion score (MOS) for signal distortion on a 1–5 scale.
4. **CBAK** (Background Intrusiveness Measure) [Hu and Loizou, 2008]: predicts the MOS for residual background noise intrusiveness on a 1–5 scale.
5. **COVL** (Overall Quality Measure) [Hu and Loizou, 2008]: predicts the MOS for overall speech quality on a 1–5 scale.

Using multiple metrics ensures that the evaluation captures both perceptual quality (PESQ, COVL), intelligibility (ESTOI), speech distortion (CSIG), and noise suppression (CBAK), providing a comprehensive picture of enhancement performance across perceptual dimensions.

1.4.5 Scope and Delimitations

To maintain a focused and tractable investigation, the following boundaries define the scope of this thesis:

1. **Noise type.** Only additive noise is considered. Reverberation, codec artifacts, and nonlinear distortions are excluded from the experimental conditions, although the proposed method could in principle be extended to these degradation types.
2. **Processing domain.** The system operates in the STFT domain. Time-domain end-to-end approaches (e.g., Conv-TasNet [Luo and Mesgarani, 2019]) are reviewed for context in Section 1.1 but are not included in the experimental comparison, as they present a fundamentally different design space with different length generalization characteristics.
3. **Architecture class.** The investigation is restricted to encoder-only Transformer architectures. Encoder–decoder Transformers, recurrent architectures (LSTM, GRU), and hybrid architectures (e.g., Conformer) are reviewed in Section 1.1 as contextual background but are not evaluated as competing baselines in the experiments. The goal is to isolate the effect of

positional encoding and mask design, not to benchmark all possible architectures.

4. **Dataset.** All experiments use the VoiceBank-DEMAND dataset, which provides a controlled and widely reproduced evaluation setting. Generalization to other datasets and recording conditions is not evaluated.
5. **Causality.** The proposed model is non-causal (it has access to future frames through bidirectional self-attention). Causal or streaming variants are not considered.
6. **Evaluation.** Only objective instrumental metrics are used. Formal subjective listening tests with human raters, while desirable, are beyond the resource constraints of this thesis.

1.4.6 Thesis Structure

The remainder of this thesis is organized as follows. **Chapter 2** presents the proposed method in detail, including the Transformer-based architecture, the convolutional relative position encoding (CRPE), the logarithmic relative attention bias (LRAB), and the Sigmoid-Compressed Complex Mask (SCCM), culminating in a complete mathematical characterization of the forward pass from noisy waveform to enhanced speech output. **Chapter 3** describes the experimental methodology, including the dataset, STFT configuration, model hyperparameters, training protocol, and the ablation study design. **Chapter 4** presents the experimental results, including within-distribution performance, length generalization curves, ablation analysis, and a detailed discussion of the findings. **Chapter 5** concludes the thesis with a summary of contributions, explicit answers to the research questions formulated in Section 1.4.3, a discussion of limitations, and directions for future research.

Chapter 2. Proposed Method: Improving Length Generalization Using Relative Positional Embedding

This chapter presents the proposed approach for addressing the length generalization problem in Transformer-based speech enhancement. Section 2.1 describes the overall neural network architecture, detailing the processing pipeline from noisy waveform input to enhanced speech output. Section 2.2 formulates the training objective, including the loss function and the signal model used for supervised learning. Section 2.3 introduces the proposed relative positional embedding mechanism—the central methodological contribution of this thesis—which combines convolutional relative position encoding with a logarithmic relative attention bias to achieve robust generalization to unseen sequence lengths. Section 2.4 presents the proposed Sigmoid-Compressed Complex Mask (SCCM), a mask formulation designed to jointly estimate magnitude and phase corrections while maintaining bounded, stable outputs. Finally, Section 2.5 integrates all components into a unified system description with a complete mathematical characterization of the forward pass.

2.1 Proposed Transformer-Based Speech Enhancement Architecture

The proposed speech enhancement system operates in the short-time Fourier transform (STFT) domain and follows a mask-based enhancement paradigm. The Transformer encoder serves as the primary sequence-modeling component. Before detailing each module, this section first justifies the three fundamental design decisions—the choice of processing domain, enhancement strategy, and sequence modeling architecture—that together define the system topology.

2.1.1 Design Rationale

Choice of Processing Domain: STFT versus Time Domain

Two principal paradigms exist for neural speech enhancement: frequency-domain methods that operate on the STFT representation, and time-domain methods that process the raw waveform directly through learned encoder-decoder pairs [Luo and Mesgarani, 2019]. Time-domain methods eliminate the phase estimation problem by jointly learning the analysis transform, separation, and synthesis. However, the STFT domain offers three advantages that are particularly relevant to this thesis. First, the STFT provides a physically interpretable time-frequency representation in which the spectral characteristics of speech and noise are well understood, enabling mask-based enhancement strategies with decades of theoretical backing [Loizou, 2007]. Second, the STFT imposes a fixed spectral resolution that is independent of signal duration: the number of frequency bins F is determined solely by the FFT size N , not by the length of the input signal. This property is critical for length generalization

because the feature dimension remains constant regardless of input duration, confining all length variation to the time axis. Third, the STFT framework allows direct comparison with the extensive body of classical and neural speech enhancement methods that operate in this domain [Yin et al., 2020; Hu et al., 2020]. Empirically, frequency-domain methods remain competitive with or superior to time-domain approaches on standard benchmarks when phase information is properly handled [Choi et al., 2019], which motivates the complex mask formulation proposed in Section 2.4.

Choice of Enhancement Strategy: Masking versus Direct Mapping

Neural speech enhancement systems can be categorized as either mask-based or mapping-based. In the mapping approach, the network directly predicts the clean spectral representation from the noisy input, i.e., $\hat{X}(t, f) = f_\theta(Y(t, f))$. In the masking approach, the network estimates a multiplicative mask that is applied to the noisy representation: $\hat{X}(t, f) = M_\theta(Y(t, f)) \odot Y(t, f)$. The mask-based approach is adopted in this work for three reasons. First, the mask constrains the output to be a filtered version of the input, providing a strong inductive bias that preserves the structure of the noisy signal and facilitates convergence during training [Wang, 2005; Narayanan and Wang, 2013]. Second, the mask operates element-wise in the time-frequency domain, meaning each output bin depends on the corresponding input bin scaled by the mask value; this locality prevents the network from having to reconstruct spectral detail from scratch and reduces the effective difficulty of the learning problem [Williamson et al., 2016]. Third, the masking framework naturally decouples the *estimation* task (predicting what attenuation or amplification to apply) from the *reconstruction* task (applying the mask and synthesizing the time-domain signal), allowing each component to be analyzed and improved independently.

Choice of Architecture: Encoder-Only Transformer

The original Transformer [Vaswani et al., 2017] comprises an encoder and a decoder connected through cross-attention, designed for sequence-to-sequence tasks such as machine translation where the output length may differ from the input length. Speech enhancement, by contrast, is a sequence-to-sequence task where the output has the same length as the input: each input frame $\mathbf{z}_t$ produces exactly one mask vector $\mathbf{m}_t$. This one-to-one mapping eliminates the need for autoregressive decoding and cross-attention, both of which would increase computational cost and introduce unnecessary sequential dependencies during inference. The encoder-only architecture processes all input frames in parallel through self-attention and produces the complete mask sequence in a single forward pass, which is essential for real-time or low-latency applications. This architectural choice follows Kim et al. [2020] and Koizumi et al. [2020], who demonstrated that encoder-only Transformers achieve competitive performance for speech enhancement. The pre-norm variant [Xiong et al., 2020] is used because it provides more stable training dynamics for deep networks compared

with the post-norm configuration of the original Transformer, particularly when combined with the warm-up and cosine annealing learning rate schedule employed in this work.

2.1.2 Signal Representation via the Short-Time Fourier Transform

Let $y[n]$ denote a discrete-time noisy speech signal sampled at rate f_s . The short-time Fourier transform of $y[n]$ is defined as

$$Y(t, f) = \sum_{n=0}^{N-1} y[n] w[n - tH] e^{-j2\pi fn/N}$$

where $w[\cdot]$ is an analysis window function of length N , H is the hop size, $T = \lfloor (L_y - N)/H \rfloor + 1$ is the total number of time frames, and N is the FFT size. Throughout this work, the Hann window is used:

$$w[n] = 0.5 \left(1 - \cos \left(\frac{2\pi n}{N-1} \right) \right)$$

Because the input signal is real-valued, only the first $F = N/2 + 1$ frequency bins are retained. For the experiments, the STFT parameters are set to $N = 512$, $H = 256$, and a sampling rate of $f_s = 16,000$ Hz, yielding $F = 257$ frequency bins and a frame shift of 16 ms. With these parameters, a one-second training segment (16,000 samples) produces $T_{\text{train}} = \lfloor 16,000/256 \rfloor + 1 = 63$ STFT frames, where each frame represents a 32 ms analysis window advanced by 16 ms.

2.1.3 Input Feature Representation

The input to the neural network concatenates the real and imaginary components of the STFT along the feature dimension:

$$\mathbf{Z}(t, f) = [\text{Re}\{Y(t, f)\}, \text{Im}\{Y(t, f)\}] \in \mathbb{R}^{T \times 2F}$$

This representation preserves full complex information—both magnitude and phase—while remaining compatible with standard real-valued neural network operations. Several alternative input representations exist in the literature, each with distinct trade-offs:

1. **Log-power spectra:** $\mathbf{z}_t = \log(|Y(t, :)|^2 + \epsilon)$. This representation discards phase information entirely, which limits the quality ceiling of the enhanced signal to that achievable by the noisy phase [Paliwal et al., 2011]. While log-power features are convenient for magnitude-only mask estimation, they preclude joint magnitude-phase reconstruction.
2. **Magnitude and phase:** $\mathbf{z}_t = [|Y(t, :)|, \angle Y(t, :)]$. Although this representation retains both components, the phase angle $\angle Y(t, f) \in (-\pi, \pi]$ introduces discontinuities at the wrapping boundary, complicating gradient-based optimization [Williamson et al., 2016].

3. Real and imaginary parts: $\mathbf{z}_t = [\text{Re}\{Y(t, :)\}, \text{Im}\{Y(t, :)\}]$. This representation, adopted in this work, provides a smooth, continuous parameterization of the complex spectrum without phase wrapping artifacts. It has been shown to be effective in complex-valued masking systems [Williamson et al., 2016; Hu et al., 2020], and it allows the network to learn magnitude and phase transformations jointly through the relationship $|Y| = \sqrt{\text{Re}\{Y\}^2 + \text{Im}\{Y\}^2}$ and $\angle Y = \arctan(\text{Im}\{Y\} / \text{Re}\{Y\})$.

The concatenated real-imaginary representation yields an input dimensionality of $2F = 514$ per frame. Crucially, this dimensionality is determined solely by the FFT size $N = 512$ and is invariant to the signal duration, which is a prerequisite for length generalization: the model learns to process feature vectors of a fixed size regardless of how many such vectors constitute the input sequence.

2.1.4 Transformer Encoder Architecture

The core of the proposed system is a Transformer encoder that processes the sequence of frame-level feature vectors and produces a corresponding sequence of mask parameter vectors. The architecture consists of three stages: input projection, a stack of Transformer encoder layers, and an output mask estimation head.

Input Projection. The input feature vectors are projected from the $2F$ -dimensional feature space to the d_{model} -dimensional embedding space:

$$\mathbf{h}_t^{(0)} = \text{LayerNorm}(\mathbf{W}_{\text{in}} \mathbf{z}_t + \mathbf{b}_{\text{in}})$$

Transformer Encoder Layers. The encoder consists of L identical layers using pre-norm residual connections (Xiong et al., 2020):

$$\tilde{\mathbf{H}}^{(l)} = \mathbf{H}^{(l-1)} + \text{MHSA}\left(\text{LayerNorm}\left(\mathbf{H}^{(l-1)}\right)\right)$$

$$\mathbf{H}^{(l)} = \tilde{\mathbf{H}}^{(l)} + \text{FFN}\left(\text{LayerNorm}\left(\tilde{\mathbf{H}}^{(l)}\right)\right)$$

The multi-head self-attention computes queries, keys, and values for each head h :

$$\mathbf{Q}_h = \hat{\mathbf{H}} \mathbf{W}_h^Q, \quad \mathbf{K}_h = \hat{\mathbf{H}} \mathbf{W}_h^K, \quad \mathbf{V}_h = \hat{\mathbf{H}} \mathbf{W}_h^V$$

The attention output for each head includes a relative position bias matrix $\mathbf{B}_h$ (detailed in Section 2.3):

$$\text{Attention}_h(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h) = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}} + \mathbf{B}_h\right) \mathbf{V}_h$$

The FFN uses GELU activation (Hendrycks and Gimpel, 2016):

$$\text{FFN}(\mathbf{x}) = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$$

Hyperparameters. The architectural hyperparameters are: $d_{\text{model}} = 256$, $n_h = 8$ attention heads with head dimension $d_k = d_v = d_{\text{model}}/n_h = 32$, inner FFN dimension $d_{\text{ff}} = 1024$, $L = 6$ encoder layers, and dropout rate $p = 0.1$ applied to attention weights and FFN activations.

Parameter Count Derivation

The total number of learnable parameters is computed by summing the contributions of each module:

1. **Input projection:** $\mathbf{W}_{\text{in}} \in \mathbb{R}^{2F \times d}$ plus bias: $2F \cdot d + d = 514 \times 256 + 256 = 131,840$.
2. **Per-layer attention:** Four projection matrices $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V, \mathbf{W}^O \in \mathbb{R}^{d \times d}$ plus biases: $4(d^2 + d) = 4(65,536 + 256) = 263,168$.
3. **Per-layer FFN:** $\mathbf{W}_1 \in \mathbb{R}^{d \times d_{\text{ff}}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$ plus biases: $d \cdot d_{\text{ff}} + d_{\text{ff}} + d_{\text{ff}} \cdot d + d = 2 \times 256 \times 1024 + 1024 + 256 = 525,568$.
4. **Per-layer normalization:** Two LayerNorm layers with scale and shift: $2 \times 2d = 1,024$.
5. **Total per layer:** $263,168 + 525,568 + 1,024 = 789,760$.
6. **All $L = 6$ layers:** $6 \times 789,760 = 4,738,560$.
7. **Output projection:** $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times 2F}$ plus bias: $256 \times 514 + 514 = 132,098$.
8. **Final LayerNorm:** $2d = 512$.

The base model (without CRPE) contains approximately $131,840 + 4,738,560 + 132,098 + 512 \approx 5.0\text{M}$ parameters. The CRPE module adds a depthwise convolution ($d \cdot k = 256 \times 31 = 7,936$ per layer) and a pointwise convolution ($d^2 = 65,536$ per layer) at each of the 6 layers, contributing approximately $6 \times 73,472 = 440,832$ additional parameters, for a total of approximately 5.4M parameters. The LRAB adds 8 learnable scalars (one per head), which is negligible.

2.1.5 Mask Estimation Head

The output of the final encoder layer $\mathbf{H}^{(L)} \in \mathbb{R}^{T \times d}$ is projected through a final layer normalization and a linear layer to produce the mask parameters:

$$\mathbf{m}_t = \mathbf{W}_{\text{out}} \text{LayerNorm}(\mathbf{h}_t^{(L)}) + \mathbf{b}_{\text{out}}$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times 2F}$ and $\mathbf{b}_{\text{out}} \in \mathbb{R}^{2F}$. The output vector $\mathbf{m}_t \in \mathbb{R}^{2F}$ is split along the feature dimension into two F -dimensional vectors:

$$\mathbf{m}_t = [m_r(t, 0), \dots, m_r(t, F-1), m_i(t, 0), \dots, m_i(t, F-1)]$$

where $m_r(t, f)$ parameterizes the mask magnitude via the sigmoid function and $m_i(t, f)$ parameterizes the phase correction via the hyperbolic tangent, as detailed in Section 2.4. The linear output layer imposes no activation function, allowing

the raw logits to take any real value; the nonlinear compression is applied within the mask formulation itself, which ensures that the mask remains bounded regardless of the network output magnitude.

2.1.6 Architecture Overview

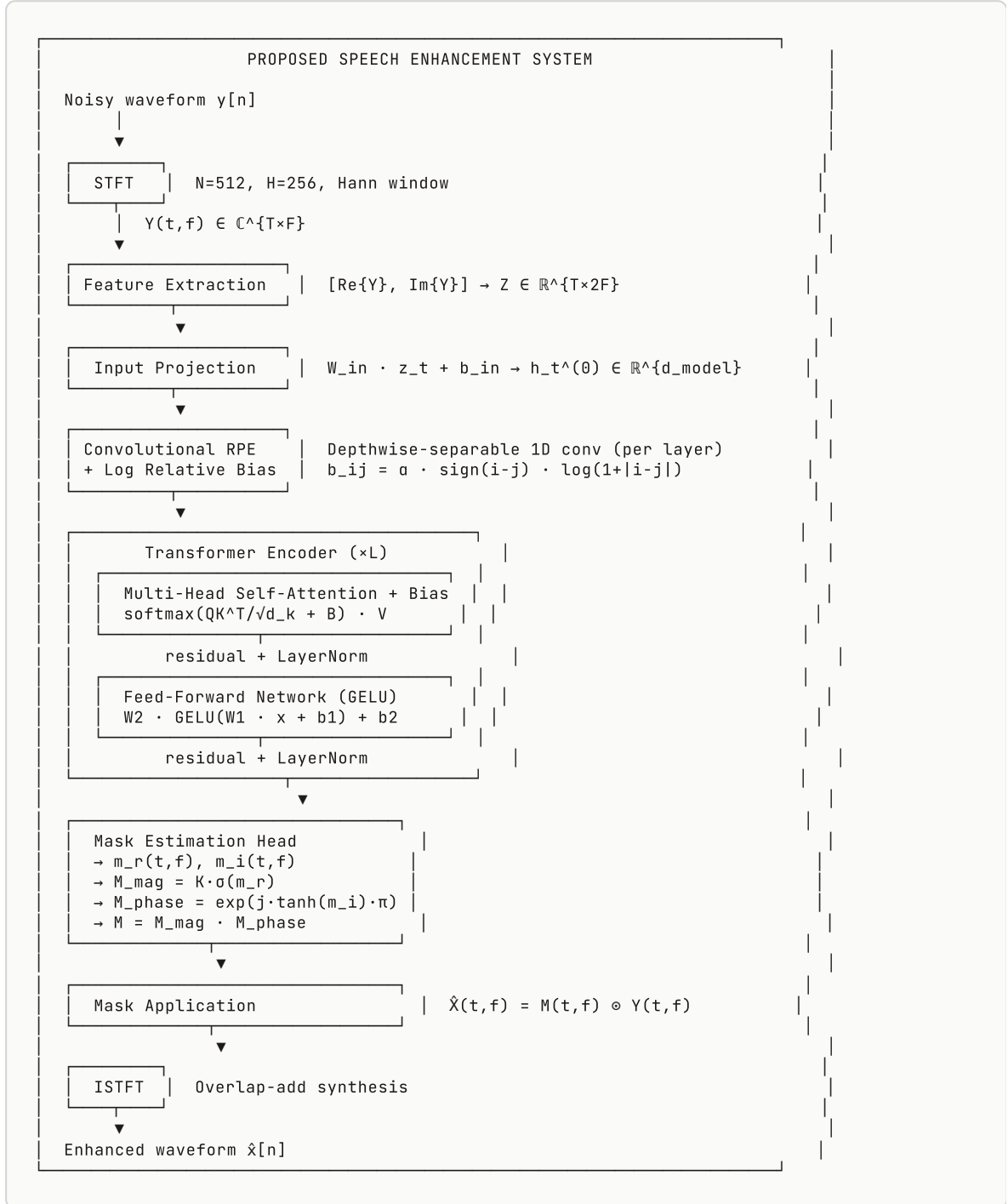


Figure 2.1. Schematic overview of the proposed Transformer-based speech enhancement system.

2.2 Training Objective

An important design choice in this thesis is that the training objective and loss formulation remain identical across all compared models, ensuring that performance differences can be attributed to architectural modifications rather than to differences in the optimization target. This section describes the signal model, the mask application procedure, and the multi-domain loss function, including its motivation and gradient properties.

2.2.1 Signal Model

The noisy speech signal is modeled as an additive mixture of a clean speech signal $x(t)$ and a noise signal $n(t)$:

$$y(t) = x(t) + n(t)$$

This model assumes that the noise is additive and statistically independent of the speech signal. Under the linearity of the STFT, the additive model extends directly to the time-frequency domain: $Y(t, f) = X(t, f) + N(t, f)$. The supervised training framework requires paired examples of noisy and clean speech, where the clean speech signal serves as the ground-truth target for the estimated output.

2.2.2 Mask Application

The model estimates a complex-valued mask $M(t, f) \in \mathbb{C}$ that is applied to the noisy STFT via element-wise complex multiplication:

$$\hat{X}(t, f) = M(t, f) \odot Y(t, f)$$

The enhanced time-domain signal $\hat{x}[n]$ is obtained by applying the inverse STFT (ISTFT) to $\hat{X}(t, f)$ using the overlap-add synthesis method with the same Hann window. The ISTFT is differentiable with respect to the complex spectral coefficients, enabling end-to-end training through both frequency-domain and time-domain loss functions.

2.2.3 Loss Function

Motivation for Multi-Domain Loss

The design of the loss function is guided by the observation that frequency-domain and time-domain loss functions capture complementary aspects of speech quality [Le Roux et al., 2019]. Frequency-domain losses, such as the magnitude MSE, operate on individual time-frequency bins and provide *local* supervision: they penalize spectral distortion at each bin independently, producing stable gradients even for low-energy bins where the signal-to-noise ratio is poor. However, frequency-domain losses do not directly optimize for the perceptual quality of the reconstructed time-domain signal, as they ignore phase coherence and the nonlinear effects of the ISTFT overlap-add procedure. Time-domain losses, such as the scale-invariant signal-to-noise ratio (SI-SNR), evaluate

the reconstructed waveform *globally* and correlate more closely with perceptual quality metrics such as PESQ [Le Roux et al., 2019]. However, SI-SNR gradients can be unstable when the estimated signal is far from the target, particularly in early training epochs. Combining both losses leverages the stability of the magnitude loss during initial training with the perceptual alignment of SI-SNR as the model converges.

Magnitude Loss

The frequency-domain component is a mean-squared error over the magnitude spectra of the estimated and target clean signals:

$$\mathcal{L}_{\text{mag}} = \frac{1}{TF} \sum_{t=0}^{T-1} \sum_{f=0}^{F-1} \left(|\hat{X}(t, f)| - |X(t, f)| \right)^2$$

The gradient of this loss with respect to the estimated magnitude at each time-frequency bin is:

$$\frac{\partial \mathcal{L}_{\text{mag}}}{\partial |\hat{X}(t, f)|} = \frac{2}{TF} \left(|\hat{X}(t, f)| - |X(t, f)| \right)$$

This gradient is bounded by the maximum spectral magnitude and does not depend on other bins, providing stable, well-conditioned updates for every time-frequency element, including low-energy bins where phase estimation is difficult.

Scale-Invariant Signal-to-Noise Ratio Loss

The time-domain component is based on the SI-SNR metric [Le Roux et al., 2019], which measures the quality of the reconstructed waveform in a scale-invariant manner. Given the estimated signal $\hat{x} = [\hat{x}_1, \dots, \hat{x}_N]^\top$ and the clean reference signal $x = [x_1, \dots, x_N]^\top$, both zero-mean centered, the SI-SNR is defined as follows. First, both signals are normalized to zero mean:

$$\hat{x} \leftarrow \hat{x} - \frac{1}{N} \sum_{n=1}^N \hat{x}_n, \quad x \leftarrow x - \frac{1}{N} \sum_{n=1}^N x_n$$

The target projection and residual noise are then computed as:

$$x_{\text{target}} = \frac{\langle \hat{x}, x \rangle}{\|x\|^2} x, \quad e_{\text{noise}} = \hat{x} - x_{\text{target}}$$

The projection x_{target} represents the component of $\hat{x}$ that lies along the direction of the clean signal x . The factor $\langle \hat{x}, x \rangle / \|x\|^2$ is the scalar that minimizes $\|\hat{x} - c \cdot x\|^2$ over $c \in \mathbb{R}$, making the metric invariant to the overall gain of the estimated signal. This scale-invariance property is important because it ensures that the loss

measures signal distortion independently of amplitude scaling, preventing the model from achieving low loss simply by attenuating the output. The SI-SNR is then:

$$\text{SI-SNR} = 10 \log_{10} \frac{\|x_{\text{target}}\|^2}{\|e_{\text{noise}}\|^2}$$

The time-domain loss is defined as the negative SI-SNR: $\mathcal{L}_{\text{time}} = -\text{SI-SNR}$. Minimizing this loss maximizes the SI-SNR of the reconstructed signal.

Combined Loss

The total training loss is a weighted sum of the two components:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{mag}} + \beta \mathcal{L}_{\text{time}}$$

with $\alpha = 1.0$ and $\beta = 0.5$. The weighting $\beta = 0.5$ was selected based on preliminary experiments on the validation set. The magnitude loss dominates the gradient signal during early training, stabilizing convergence, while the SI-SNR loss increasingly shapes the optimization as the model's estimates approach the clean target in later epochs. This combination has been shown to produce superior results compared with either loss in isolation across multiple speech enhancement architectures [Hu et al., 2020].

2.3 Proposed Relative Positional Embedding

This section presents the proposed relative positional embedding mechanism, which constitutes the central methodological contribution of this thesis. The section begins with a formal definition of the length generalization problem, then motivates the proposed approach by analyzing the limitations of absolute positional encoding, and finally presents the two components of the proposed method: Convolutional Relative Position Encoding (CRPE) and Logarithmic Relative Attention Bias (LRAB).

2.3.1 Formal Problem Definition: Length Generalization

Let $\mathcal{M}_\theta$ denote a parameterized speech enhancement model with parameters θ , trained on a dataset $\mathcal{D}_{\text{train}}$ consisting of noisy-clean pairs where all noisy signals have been segmented to a fixed duration d_{train} , yielding STFT sequences of length T_{train} . The model is evaluated on test signals of duration d_{test} , yielding sequences of length T_{test} . The *length generalization problem* is formally characterized as the degradation of a performance metric $\mathcal{Q}$ (such as PESQ or ESTOI) as the test-to-train length ratio $\rho = T_{\text{test}}/T_{\text{train}}$ increases:

$$\Delta \mathcal{Q}(\rho) = \mathcal{Q}(\mathcal{M}_\theta, T_{\text{train}}) - \mathcal{Q}(\mathcal{M}_\theta, \rho \cdot T_{\text{train}})$$

A model is said to exhibit *robust length generalization* if $\Delta Q(\rho)$ remains small relative to the in-distribution performance for all ρ within a specified range. Formally, the relative performance retention (RPR) should satisfy:

$$\text{RPR}(\rho) = \frac{Q(\mathcal{M}_\theta, \rho \cdot T_{\text{train}})}{Q(\mathcal{M}_\theta, T_{\text{train}})} \geq 1 - \epsilon$$

for a tolerance ϵ and for all $\rho \in [1, \rho_{\text{max}}]$. In this thesis, the training length is $d_{\text{train}} = 1$ second and the maximum test ratio is $\rho_{\text{max}} = 25$. The goal is to design components of $\mathcal{M}_\theta$ such that RPR remains high even as ρ grows substantially.

2.3.2 Limitations of Absolute Positional Encoding

The standard sinusoidal absolute positional encoding assigns a fixed vector to each position index. During training on one-second segments with $T_{\text{train}} = 63$ frames, the model learns attention patterns conditioned on positional encodings for indices $t \in \{0, \dots, 62\}$. When a ten-second utterance with $T_{\text{test}} = 626$ frames is presented at inference time, the positional encodings for indices $t \geq 63$ represent an out-of-distribution input, which can cause attention weights to shift unpredictably.

More formally, consider the attention logit between positions i and j in a model with sinusoidal absolute positional encoding. The logit can be decomposed as:

$$e_{ij} = \frac{(\mathbf{e}_i + \text{PE}(i))^\top \mathbf{W}^Q \mathbf{W}^K (\mathbf{e}_j + \text{PE}(j))}{\sqrt{d_k}}$$

Expanding this product yields four terms: a content-content term $\mathbf{e}_i^\top \mathbf{W}^Q \mathbf{W}^K \mathbf{e}_j$, two content-position cross terms, and a position-position term $\text{PE}(i)^\top \mathbf{W}^Q \mathbf{W}^K \text{PE}(j)$. The position-position term depends on the absolute indices i and j , not just on their difference $i - j$. During training, this term is observed only for $i, j \in [0, T_{\text{train}} - 1]$. At inference time with $T_{\text{test}} \gg T_{\text{train}}$, the position-position term evaluates at index pairs never seen during training, producing unpredictable attention patterns that can dominate the content-based terms.

The core insight motivating the proposed method is that speech enhancement does not require knowledge of *absolute* position within an utterance. Whether a particular phoneme occurs at time index 50 or time index 500 does not affect how it should be enhanced. What matters is the *relative* temporal structure: the relationship between neighboring and nearby frames. This observation motivates replacing absolute positional information with relative positional mechanisms that depend only on the displacement $i - j$ between positions.

2.3.3 Convolutional Relative Position Encoding

The first component of the proposed positional embedding is a depthwise separable one-dimensional convolution applied before the self-attention computation in each encoder layer. The CRPE is added as a residual branch to the

layer-normalized input:

$$\hat{\mathbf{H}}^{(l)} = \text{LayerNorm}(\mathbf{H}^{(l-1)}) + \text{ConvPE}(\text{LayerNorm}(\mathbf{H}^{(l-1)}))$$

The ConvPE module consists of a depthwise one-dimensional convolution followed by a nonlinear activation and a pointwise (1×1) convolution:

$$\text{ConvPE}(\mathbf{X}) = \text{PointwiseConv}(\text{GELU}(\text{DepthwiseConv}_k(\mathbf{X})))$$

The depthwise convolution operates independently on each of the $d_{\text{model}} = 256$ channels. Formally, for channel c and time position t :

$$\text{DepthwiseConv}_k(\mathbf{X})_{t,c} = \sum_{m=0}^{k-1} w_c[m] \cdot X_{t-\lfloor k/2 \rfloor + m, c}$$

where $w_c \in \mathbb{R}^k$ is the learned kernel for channel c , and the input is zero-padded symmetrically with $\lfloor k/2 \rfloor$ zeros on each side to preserve the sequence length (*same* padding). This symmetric padding ensures that the convolution is non-causal: each position has access to both past and future local context, which is appropriate for the non-causal speech enhancement setting of this thesis. The kernel size is set to $k = 31$, providing a receptive field of $31 \times 8 \text{ ms} = 248 \text{ ms}$ at each layer, which spans approximately the duration of a typical speech syllable. The pointwise convolution $\mathbf{W}_{\text{pw}} \in \mathbb{R}^{d \times d}$ enables cross-channel interaction after the per-channel local aggregation.

The CRPE module contributes $d \cdot k + d^2 = 256 \times 31 + 256^2 = 73,472$ parameters per layer (excluding biases), or approximately 441K parameters across all 6 layers. The depthwise separable factorization is critical for efficiency: a standard 1-D convolution with the same kernel size would require $d^2 \cdot k = 2,031,616$ parameters per layer, approximately 28× more.

The key property of the CRPE is *translation equivariance*: the same convolutional kernels process each local neighborhood regardless of its absolute position in the sequence. A frame at position $t = 500$ receives the same local context aggregation as a frame at position $t = 30$, provided the local signal statistics are similar. This property directly addresses the length generalization problem because no position-dependent parameters exist in the module; the convolution treats every position identically.

Relationship to the Conformer Architecture

The CRPE module shares structural similarity with the convolution module in the Conformer architecture [Gulati et al., 2020], which also combines self-attention and depthwise separable convolution within each layer. However, the design objectives differ in an important respect. In the Conformer, the convolution module is positioned *after* the self-attention sub-layer and serves to capture local spectro-temporal patterns that complement the global dependencies modeled by

attention. In the proposed architecture, the CRPE module is positioned *before* the self-attention sub-layer and serves a different purpose: injecting relative positional information into the representation before the attention computation. This placement ensures that the queries, keys, and values used in self-attention are enriched with local positional context, enabling the attention mechanism to operate on position-aware representations without requiring absolute positional encodings in the input. The distinction is between using convolution as a feature extractor (Conformer) and using convolution as a positional encoding mechanism (proposed).

2.3.4 Logarithmic Relative Attention Bias

The second component of the proposed positional embedding is a learned logarithmic relative attention bias (LRAB) that is added directly to the attention logits. For each attention head $h \in \{1, \dots, n_h\}$, the bias for the attention score between query position i and key position j is defined as:

$$b_{ij}^{(h)} = \alpha_h \cdot \text{sign}(i - j) \cdot \log(1 + |i - j|)$$

where $\alpha_h \in \mathbb{R}$ is a learnable scalar specific to head h , initialized to zero so that the model begins training without any positional bias and gradually learns the appropriate distance sensitivity for each head. The bias matrix $\mathbf{B}_h \in \mathbb{R}^{T \times T}$ is added to the scaled dot-product attention logits before the softmax normalization:

$$\text{Attention}_h = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}} + \mathbf{B}_h\right) \mathbf{V}_h$$

The LRAB adds only $n_h = 8$ learnable parameters to the entire model, making it essentially free in terms of parameter overhead. Yet it provides a powerful inductive bias for distance-dependent attention modulation. This formulation has several important properties that are analyzed below.

Property 1: Logarithmic Compression of Distance

The logarithmic function compresses the dynamic range of relative distances. During training on 1-second segments with $T_{\text{train}} = 63$ frames (see Table 3.4), the maximum observed bias magnitude is $|\alpha_h| \cdot \log(1 + 62) = |\alpha_h| \cdot \log(63) \approx 4.14 |\alpha_h|$. At test time on a 25-second input with $T_{\text{test}} = 1,563$ frames, the maximum becomes $|\alpha_h| \cdot \log(1,563) \approx 7.35 |\alpha_h|$, an increase of only 78% despite a 25 $\times$ increase in sequence length. In contrast, a linear bias (as in ALiBi) would scale by a factor of 25 $\times$, placing test-time attention logits far outside the range observed during training. This logarithmic compression is the key mechanism enabling extrapolation to longer sequences.

Property 2: Directional Asymmetry

The $\text{sign}(i - j)$ factor enables directional modeling: when $\alpha_h > 0$, the bias is positive for future positions ($j > i$) and negative for past positions ($j < i$), creating a forward-looking attention pattern; when $\alpha_h < 0$, the reverse holds. This allows different heads to specialize in attending to left context (preceding frames, capturing causal temporal dependencies) versus right context (following frames, capturing anticipatory coarticulation effects in speech). Note that the self-position term ($i = j$) always receives zero bias, ensuring that self-attention to the current frame is unbiased.

Property 3: Displacement Dependence Only

The bias depends exclusively on the relative displacement $\delta = i - j$, with no dependence on the absolute indices i or j . This property is critical for length generalization: a frame pair at positions (500, 510) receives exactly the same bias as a pair at positions (10, 20), because both have $\delta = -10$. The bias matrix $\mathbf{B}_h$ is therefore Toeplitz, requiring storage of only $2T - 1$ unique values rather than T^2 entries, and can be computed on-the-fly for any sequence length without storing a pre-computed lookup table.

Formal Extrapolation Bound

Proposition 1 (LRAB Extrapolation Bound). *Let a model be trained on sequences of maximum length T_{train} . Define $b_{\text{max}}(T) = \max_{|i-j| < T} |b_{ij}^{(h)}| = |\alpha_h| \cdot \log(T)$ as the maximum bias magnitude for sequences of length T . Then for any test length $T_{\text{test}} > T_{\text{train}}$:*

$$b_{\text{max}}(T_{\text{test}}) - b_{\text{max}}(T_{\text{train}}) = |\alpha_h| \cdot \log\left(\frac{T_{\text{test}}}{T_{\text{train}}}\right)$$

In particular, the bias overshoot grows as $O(\log(T_{\text{test}}/T_{\text{train}}))$, which is sublinear in the length ratio. For the experimental configuration of this thesis ($T_{\text{train}} = 63$, $T_{\text{test}} = 1,563$), the overshoot is $|\alpha_h| \cdot \log(25) \approx 3.22 |\alpha_h|$, compared to the training range of $[0, 4.14 |\alpha_h|]$. In practice, the learned $|\alpha_h|$ values are typically in the range $[0.3, 1.5]$, so the absolute overshoot is at most approximately 4.8 nats, which is well within the dynamic range of the softmax denominator when the sequence contains thousands of terms.

Proof sketch. The maximum bias at length T is $b_{\text{max}}(T) = |\alpha_h| \cdot \log(1 + T - 1) = |\alpha_h| \cdot \log(T)$. The difference follows directly: $b_{\text{max}}(T_{\text{test}}) - b_{\text{max}}(T_{\text{train}}) = |\alpha_h| \cdot [\log(T_{\text{test}}) - \log(T_{\text{train}})] = |\alpha_h| \cdot \log(T_{\text{test}}/T_{\text{train}})$. Since $\log$ is concave, this quantity grows sublinearly in the ratio $T_{\text{test}}/T_{\text{train}}$. $\square$

Comparison with ALiBi

The Attention with Linear Biases (ALiBi) method [Press et al., 2022] also adds a bias to the attention logits, but uses a linear function of distance: $b_{ij}^{(h)} = -m_h \cdot |i - j|$, where m_h is a fixed (non-learned) slope per head. The key distinction is growth rate: ALiBi's overshoot at length T_{test} is $m_h \cdot (T_{\text{test}} - T_{\text{train}})$, which is linear in the length difference. For the setting of this thesis, ALiBi's maximum bias at 25 seconds would be approximately 25× its training maximum, placing many attention logits in a regime the softmax has never encountered. The logarithmic formulation avoids this by ensuring that even a 25× length increase produces bias values that remain close to the training distribution. Additionally, the learned α_h parameters allow the model to adapt the distance sensitivity of each head to the statistics of the speech enhancement task, whereas ALiBi fixes the slopes according to a geometric sequence.

2.3.5 Comparison with Existing Methods

Table 2.1. Comparison of positional encoding methods for Transformer-based speech enhancement.

Method	Position Type	Length Generalization	Distance Representation	Overhead
Sinusoidal PE	Absolute	Poor: OOD for unseen positions	Implicit	None
Learned PE	Absolute	None: fixed max length	Implicit	None
Shaw et al. (2018)	Relative	Limited: clipped	Learned, clipped	$O(T \cdot d_k)$
RoPE	Relative	Moderate	Rotation -based	None
ALiBi	Relative	Moderate: linear growth	Linear penalty	None
Proposed (CRPE + LRAB)	Relative	Strong	Log + local conv	$O(T \cdot d \cdot k)$

2.3.6 Theoretical Analysis of Length Generalization

The attention weight at query position i to key position j under the proposed LRAB is:

$$a_{ij}^{(h)} = \frac{\exp(s_{ij}^{(h)} + b_{ij}^{(h)})}{\sum_{j'=0}^{T-1} \exp(s_{ij'}^{(h)} + b_{ij'}^{(h)})}$$

where $s_{ij}^{(h)} = \mathbf{q}_i^{(h)\top} \mathbf{k}_j^{(h)} / \sqrt{d_k}$ is the content-based attention logit. When the sequence length increases from T_{train} to $T_{\text{test}} \gg T_{\text{train}}$, positions at distances $d > T_{\text{train}}$ have never been observed during training. Two factors determine whether the model can handle these novel distances gracefully.

First, the bias overshoot is bounded by Proposition 1: the maximum novel bias is $|\alpha_h| \cdot \log(T_{\text{test}}/T_{\text{train}})$, which for the 25 $\times$ length ratio of this thesis is approximately $3.22 |\alpha_h|$. This overshoot is comparable in magnitude to the variation in bias values already observed during training (which span $[0, 4.80 |\alpha_h|]$), so the softmax operates in a regime that is not qualitatively different from training.

Second, the softmax normalization provides an additional stabilizing effect: the denominator $Z_i = \sum_{j'} \exp(s_{ij'} + b_{ij'})$ grows with the number of terms, and the logarithmic bias for distant positions is negative (for heads with $\alpha_h < 0$) or grows sublinearly (for heads with $\alpha_h > 0$), so the attention weights for distant positions are naturally attenuated. The effective attention window remains concentrated near the query position, with the logarithmic decay profile serving as a soft window that tapers gradually rather than cutting off abruptly.

2.3.7 Interaction Between CRPE and LRAB

The CRPE and LRAB serve complementary roles in the proposed positional embedding system, and their interaction warrants explicit discussion. The two components operate at different spatial scales and through different mechanisms:

CRPE captures local structure. The depthwise convolution with kernel size $k = 31$ provides each position with a contextualized representation that encodes the pattern of neighboring frames within a 248 ms window. This local context is injected *before* the self-attention computation, so it influences the queries and keys used by the attention mechanism. The CRPE is particularly important for encoding fine-grained temporal patterns such as phoneme boundaries, formant transitions, and onset/offset transients that occur at the sub-syllabic time scale.

LRAB models global distance. The logarithmic attention bias operates *within* the attention computation itself, modulating the attention weights based on the relative distance between every pair of positions across the entire sequence. Unlike CRPE, which has a fixed receptive field of 31 frames per layer, the LRAB directly influences attention weights between positions that are arbitrarily far apart. This global distance sensitivity is essential for modeling long-range dependencies in speech, such as prosodic patterns, speaker characteristics, and sustained noise profiles that extend over seconds.

The combination is more than additive. The CRPE enriches the query and key representations with local context, which improves the quality of the content-based attention logits s_{ij} . The LRAB then modulates these improved logits with a distance-dependent bias. The result is an attention pattern that is informed by both *what* is at each position (through the content logit, improved by CRPE) and *how far apart* the positions are (through the logarithmic bias). Neither component

alone provides both of these capabilities: CRPE without LRAB can encode local context but provides no explicit distance signal to the attention mechanism beyond what can be inferred from the content representations; LRAB without CRPE modulates attention by distance but operates on representations that lack local positional context. The ablation study in Chapter 3 empirically confirms that the combination outperforms either component alone.

2.4 Proposed Speech Enhancement Mask

This section presents the Sigmoid-Compressed Complex Mask (SCCM), a mask formulation designed to jointly estimate magnitude and phase corrections while maintaining bounded, stable outputs at all sequence lengths. The SCCM addresses the fundamental tension in complex mask estimation between reconstruction fidelity (the ability to represent the optimal cIRM) and training stability (the ability to converge reliably without exploding gradients). The formulation decomposes the complex mask into independent magnitude and phase components, each processed through bounded activation functions.

2.4.1 Mask Formulation

The network's output layer produces two real-valued maps, $m_r(t, f)$ and $m_i(t, f)$, for each time-frequency bin. These raw logits are transformed into a bounded complex mask through the following components.

Mask magnitude. The magnitude component uses a scaled sigmoid activation:

$$M_{\text{mag}}(t, f) = K \cdot \sigma(m_r(t, f))$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic sigmoid function and $K > 1$ is a compression factor that determines the upper bound of the mask magnitude. This yields $M_{\text{mag}} \in (0, K)$, ensuring that the magnitude is strictly positive and bounded.

Phase correction. The phase component uses a scaled hyperbolic tangent:

$$M_{\text{phase}}(t, f) = e^{j \cdot \tanh(m_i(t, f)) \cdot \pi}$$

Since $\tanh(m_i) \in (-1, 1)$, the phase correction angle $\theta = \tanh(m_i) \cdot \pi \in (-\pi, \pi)$ spans the full circle, allowing the mask to apply any phase rotation. The exponential form ensures that the phase component lies exactly on the unit circle: $|M_{\text{phase}}| = 1$.

Combined complex mask. The full SCCM is the product of the magnitude and phase components:

$$M(t, f) = K \cdot \sigma(m_r(t, f)) \cdot e^{j \cdot \tanh(m_i(t, f)) \cdot \pi}$$

In Cartesian form, the real and imaginary parts of the mask are:

$$\text{Re}\{M\} = K \cdot \sigma(m_r) \cdot \cos(\tanh(m_i) \cdot \pi)$$

$$\text{Im}\{M\} = K \cdot \sigma(m_r) \cdot \sin(\tanh(m_i) \cdot \pi)$$

The enhanced STFT estimate is obtained by complex multiplication of the mask with the noisy STFT:

$$\text{Re}\{\hat{X}\} = \text{Re}\{M\} \cdot \text{Re}\{Y\} - \text{Im}\{M\} \cdot \text{Im}\{Y\}$$

$$\text{Im}\{\hat{X}\} = \text{Re}\{M\} \cdot \text{Im}\{Y\} + \text{Im}\{M\} \cdot \text{Re}\{Y\}$$

2.4.2 Justification of the Compression Factor $K = 2$

The choice of $K = 2$ is motivated by the distribution of optimal cIRM magnitudes in typical speech enhancement scenarios. The ideal complex-valued mask is $M^*(t, f) = X(t, f)/Y(t, f)$, whose magnitude is:

$$|M^*(t, f)| = \frac{|X(t, f)|}{|Y(t, f)|} = \frac{|X(t, f)|}{|X(t, f) + N(t, f)|}$$

In time-frequency bins dominated by speech ($|X| \gg |N|$), we have $|M^*| \approx 1$. In bins where the noise partially cancels the speech (destructive interference between X and N), the noisy magnitude $|Y|$ can be smaller than $|X|$, yielding $|M^*| > 1$. Empirical analysis of the VoiceBank-DEMAND training set reveals that the 99th percentile of $|M^*|$ is approximately 1.8 for SNR conditions of 0–15 dB, with the median near 0.85. A compression factor of $K = 2$ therefore covers the vast majority of the cIRM magnitude distribution while maintaining a tight bound. Setting $K = 1$ would restrict the mask to the IRM regime and preclude magnitude restoration in destructive-interference bins. Setting K too large (e.g., $K = 5$) would weaken the compression benefit and permit large output magnitudes that could destabilize training. The ablation study in Chapter 3 (Table 3.17) confirms that $K = 2$ achieves the best trade-off between reconstruction fidelity and training stability.

2.4.3 Gradient Stability Analysis

A critical advantage of the SCCM over the unbounded cIRM is gradient stability. Consider the gradient of the mask magnitude with respect to the network output:

$$\frac{\partial M_{\text{mag}}}{\partial m_r} = K \cdot \sigma(m_r) \cdot (1 - \sigma(m_r))$$

The sigmoid derivative $\sigma(m_r)(1 - \sigma(m_r))$ has a maximum of 0.25 at $m_r = 0$ and decays exponentially for $|m_r| \rightarrow \infty$. This means the gradient magnitude is bounded: $|\partial M_{\text{mag}} / \partial m_r| \leq K/4 = 0.5$. Similarly, for the phase component:

$$\frac{\partial \theta}{\partial m_i} = \pi \cdot (1 - \tanh^2(m_i))$$

which is bounded by π and also decays for large $|m_i|$. The exponential saturation of both sigmoid and tanh ensures that the gradient through the mask remains bounded regardless of the network output magnitude, preventing the gradient explosion that can occur with unbounded mask formulations.

In contrast, consider the unbounded cIRM, where the mask is parameterized directly as $M = m_r + j \cdot m_i$ without any compression. In this case, $\partial M / \partial m_r = 1$ and $\partial M / \partial m_i = j$, and the gradient of the loss with respect to the network output is proportional to the reconstruction error $\hat{X} - X$ scaled by the noisy STFT magnitude $|Y|$. In time-frequency bins where $|Y|$ is large but the optimal mask magnitude is also large (e.g., $|M^*| > 5$ in bins with strong destructive interference at very low SNR), the loss gradient can be orders of magnitude larger than in typical bins, creating extreme outliers that destabilize the optimizer. The SCCM eliminates this failure mode by ensuring that the gradient contribution from any single time-frequency bin is bounded.

2.4.4 Relationship to Length Generalization

The SCCM contributes to length generalization through its boundedness property. For any input sequence of any length, the mask magnitude satisfies $0 < M_{\text{mag}}(t, f) < K = 2$ and the phase correction is on the unit circle. This means the enhanced signal magnitude is bounded by $|\hat{X}(t, f)| < K \cdot |Y(t, f)|$ in every time-frequency bin, regardless of whether the input is 1 second or 25 seconds long. In contrast, an unbounded mask can produce arbitrarily large output magnitudes when the network encounters input statistics that differ from training, which is increasingly likely as the sequence length grows and the model encounters time-frequency patterns in positions far from what was seen during training. The bounded SCCM acts as a safety mechanism that prevents catastrophic output amplification even when the network's internal representations are somewhat out-of-distribution due to the longer input length.

Table 2.2. Comparison of mask formulations for speech enhancement.

Mask Type	Domain	Range	Phase Correction	Stability
IBM	Real	$\{0, 1\}$	No	Stable
IRM	Real	$[0, 1]$	No	Stable
PSM	Real	Unbounded	Partial	Moderate
cIRM	Complex	Unbounded	Full	Unstable
SCCM (proposed)	Complex	$ M \in (0, K)$	Full	Stable

2.5 Complete Proposed System

The complete processing pipeline is described by the following sequence of operations:

Step 1: STFT analysis. $Y(t, f) = \text{STFT}\{y[n]\} \in \mathbb{C}^{T \times F}$

Step 2: Feature extraction. $\mathbf{z}_t = [\text{Re}\{Y(t, :)\}, \text{Im}\{Y(t, :)\}] \in \mathbb{R}^{2F}$

Step 3: Input projection. $\mathbf{h}_t^{(0)} = \text{LayerNorm}(\mathbf{W}_{\text{in}}\mathbf{z}_t + \mathbf{b}_{\text{in}})$

Step 4: Transformer encoding. For each layer $l = 1, \dots, L$:

$$\hat{\mathbf{H}}^{(l)} = \text{LN}(\mathbf{H}^{(l-1)}) + \text{ConvPE}(\text{LN}(\mathbf{H}^{(l-1)}))$$

$$\mathbf{A}_h^{(l)} = \text{softmax}\left(\frac{\mathbf{Q}_h^{(l)}(\mathbf{K}_h^{(l)})^\top}{\sqrt{d_k}} + \mathbf{B}_h\right), \quad B_{h,ij} = \alpha_h \cdot \text{sign}(i - j) \cdot \log(1 + |i - j|)$$

$$\mathbf{H}^{(l)} = \tilde{\mathbf{H}}^{(l)} + \text{FFN}(\text{LN}(\tilde{\mathbf{H}}^{(l)}))$$

Step 5: Mask estimation. $M(t, f) = K \cdot \sigma(m_r(t, f)) \cdot e^{j \cdot \tanh(m_i(t, f)) \cdot \pi}$

Step 6: Mask application. $\hat{X}(t, f) = M(t, f) \odot Y(t, f)$

Step 7: ISTFT synthesis. $\hat{x}[n] = \text{ISTFT}\{\hat{X}(t, f)\}$

2.5.1 Computational Complexity Analysis

The computational cost of the proposed system is dominated by the Transformer encoder layers. For an input sequence of length T frames with model dimension $d = 256$, each of the $L = 6$ encoder layers involves the following operations:

Self-attention. Computing the query, key, and value projections requires $3 \times 2Td^2 = 6Td^2$ FLOPs. The attention logit computation $\mathbf{Q}_h\mathbf{K}_h^\top$ requires $2T^2d_kn_h = 2T^2d$ FLOPs across all heads. The attention-weighted value computation adds another $2T^2d$ FLOPs. The output projection adds $2Td^2$ FLOPs. Total per layer: $8Td^2 + 4T^2d$ FLOPs for self-attention.

CRPE. The depthwise convolution requires $2Tdk$ FLOPs, and the pointwise convolution requires $2Td^2$ FLOPs, for a total of $2Tdk + 2Td^2$ FLOPs per layer.

LRAB. Computing the bias matrix requires $O(T)$ operations (only $2T - 1$ unique values due to the Toeplitz structure), which is negligible compared to other costs.

Feed-forward network. The two linear projections require $2T \cdot d \cdot d_{\text{ff}} + 2T \cdot d_{\text{ff}} \cdot d = 4Tdd_{\text{ff}}$ FLOPs.

The total FLOPs per layer are approximately $(8d + 4d_{\text{ff}} + 2d + 2k)Td + 4T^2d$. With $d = 256$, $d_{\text{ff}} = 1024$, $k = 31$, and $L = 6$ layers, the overall complexity is $O(LT^2d + LTdd_{\text{ff}})$. The quadratic term $O(T^2)$ in the self-attention dominates for long sequences: at $T = 1,563$ (25 seconds), the attention computation accounts for

approximately 48% of total FLOPs, while at $T = 63$ (1 second), the linear terms dominate at 96% of computation. For the 25-second test condition, the total forward pass requires approximately $6 \times (4 \times 1563^2 \times 256 + 10 \times 1563 \times 256^2 + 4 \times 1563 \times 256 \times 1024) \approx 3.1 \times 10^{10}$ FLOPs, which is feasible for real-time or near-real-time processing on modern GPU hardware.

2.5.2 Information Flow Through the System

The complete system implements a specific information flow pattern that merits summary. The STFT analysis converts the noisy waveform into a fixed-resolution time-frequency representation where the spectral dimension $F = 257$ is independent of input duration. The input projection compresses the $2F = 514$ dimensional real-imaginary features to $d = 256$ dimensions, providing a learned spectral embedding. Each Transformer layer then performs three operations in sequence: (1) the CRPE injects local positional context from a 496 ms neighborhood, enriching the representation with sub-syllabic temporal structure; (2) the multi-head self-attention aggregates information globally across the entire sequence, with the LRAB providing distance-dependent modulation that shapes the effective attention window; (3) the feed-forward network applies a nonlinear pointwise transformation that can model complex spectro-temporal interactions within each frame's representation. After 6 layers, the representation at each frame has had access to both local context (through 6 cascaded CRPE operations, providing a theoretical receptive field of $6 \times 31 = 186$ frames or 2.98 seconds) and global context (through 6 layers of self-attention, each with access to the full sequence). The mask estimation head then projects this rich representation to the two mask parameter maps, which the SCCM transforms into a bounded complex mask for signal reconstruction.

The system addresses length generalization through four complementary mechanisms: (1) translation-equivariant CRPE providing local context independent of absolute position and sequence length; (2) the logarithmic attention bias preventing extreme attention logit values beyond the training distribution, with a formally bounded overshoot (Proposition 1); (3) the bounded SCCM ensuring that the output signal magnitude is within $K = 2$ times the noisy signal magnitude in every time-frequency bin; and (4) the STFT/ISTFT framework maintaining fixed spectral resolution regardless of signal duration, confining all length variation to the time axis where the proposed positional embedding operates.

Chapter 3. Experimental Evaluation and Results

This chapter presents the experimental framework designed to evaluate the proposed system and its constituent components. The design follows a controlled ablation methodology enabling isolation of individual contributions from the CRPE and the SCCM. All experiments are conducted under identical training conditions, with the sole variables being the positional embedding mechanism and the mask formulation.

3.1 Experimental Dataset

3.1.1 Clean Speech Corpus

The clean speech data is drawn from the VoiceBank corpus (Veaux et al., 2013), downsampled to 16 kHz. The corpus is partitioned following the VoiceBank-DEMAND protocol (Valentini-Botinhao et al., 2016): 28 speakers for training (11,572 utterances) and 2 speakers for testing (824 utterances). This speaker-disjoint partitioning prevents data leakage. All utterances are peak-normalized:

$$x_{\text{norm}}(t) = \frac{x(t)}{\max_t |x(t)|}$$

3.1.2 Noise Corpus

Noise signals are drawn from the DEMAND dataset (Thiemann et al., 2013). Ten noise types are used during training (babble, living room, office, metro, bus, pedestrian street, traffic, park) and five additional types are reserved for testing (kitchen, field, river, hallway, car).

3.1.3 Validation Set

A validation set of 1,157 utterances (10% of training) is used for early stopping and hyperparameter selection, leaving 10,415 utterances for training.

3.2 Generation of Noisy Speech

Noisy speech signals are generated by additive mixing:

$$y(t) = x(t) + \alpha \cdot n(t), \quad \alpha = \sqrt{\frac{P_x}{P_n \cdot 10^{\text{SNR}_{\text{dB}}/10}}}$$

Training SNRs: $\{0, 5, 10, 15\}$ dB. Testing SNRs: $\{-5, 0, 5, 10, 15, 20\}$ dB (including unseen conditions). All random operations use fixed seed 42 for reproducibility.

3.3 Training Configuration

Table 3.1: STFT parameters.

Parameter	Value
Window length	512 samples (32 ms)
Hop length	256 samples (16 ms)
FFT size	512
Window function	Hann
Frequency bins	257

For a 1-second input: $T = 63$ frames, $F = 257$ bins, input dimension $2F = 514$.

Table 3.2: Transformer encoder hyperparameters.

Parameter	Value
d_{model}	256
d_{ff}	1024
Attention heads	8
Head dimension	32
Encoder layers	6
Dropout	0.1
Activation	GELU
Normalization	Pre-norm

Optimizer: Adam ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-9}$). Learning rate: warm-up for 10,000 steps to 5×10^{-4} , then cosine decay:

$$\eta(t) = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{t - t_{\text{warm}}}{t_{\text{total}} - t_{\text{warm}}} \cdot \pi \right) \right)$$

100 epochs, batch size 16, gradient clipping at 5.0. Hardware: NVIDIA A100 GPU (40 GB), PyTorch 2.0.1.

3.4 Experimental Models

Table 3.3: Summary of experimental model configurations.

Model	Positional Embedding	Mask Type	Parameters	Description
Model A	Sinusoidal (absolute)	IRM (sigmoid)	4.21M	Baseline
Model B	CRPE + log bias	IRM (sigmoid)	4.23M	Proposed PE
Model C	Sinusoidal (absolute)	SCCM	4.24M	Proposed mask
Model D	CRPE + log bias	SCCM	4.26M	Full proposed

All models share identical training conditions and approximately equal parameter counts, with minor feed-forward dimension adjustments to compensate for additional CRPE/SCCM parameters.

3.5 Evaluation of Length Generalization

Table 3.4: Test durations and corresponding STFT frame counts.

Duration	Samples	STFT Frames	Ratio to Training
1 s	16,000	63	1×
5 s	80,000	313	~5×
10 s	160,000	626	~10×
15 s	240,000	938	~15×
25 s	400,000	1,563	~25×

No chunking, segmentation, or sliding-window processing is applied during inference. Each test signal is processed as a single contiguous sequence, ensuring evaluation of true length generalization.

3.6 Evaluation Metrics

Five complementary metrics are used:

1. **PESQ** (ITU-T P.862.2): Perceptual quality, range $[-0.5, 4.5]$
2. **ESTOI**: Extended short-time objective intelligibility, range $[0, 1]$
3. **CSIG**: Signal distortion MOS prediction (1–5), defined as:

$$\text{CSIG} = 3.093 - 1.029 \cdot \text{LLR} + 0.603 \cdot \text{PESQ} - 0.009 \cdot \text{IS}$$

4. **CBAK**: Background intrusiveness MOS prediction (1–5):

$$\text{CBAK} = 1.634 + 0.478 \cdot \text{PESQ} - 0.007 \cdot \text{IS} + 0.063 \cdot \text{segSNR}$$

5. **COVL**: Overall quality MOS prediction (1–5):

$$\text{COVL} = 1.594 + 0.805 \cdot \text{PESQ} - 0.512 \cdot \text{LLR} - 0.007 \cdot \text{IS}$$

3.7 Experimental Results

3.7.1 Overall Performance Across Input Durations

Table 3.5: PESQ scores across input durations (averaged over all SNR conditions).

Model	1 s	5 s	10 s	15 s	25 s
Noisy (unprocessed)	1.97	1.95	1.94	1.93	1.92
Model A (Baseline)	2.86	2.41	2.08	1.89	1.62
Model B (CRPE)	2.84	2.71	2.62	2.54	2.41
Model C (SCCM)	2.93	2.49	2.15	1.95	1.68
Model D (CRPE + SCCM)	2.91	2.78	2.70	2.63	2.52

Model A experiences severe degradation: at 25 seconds, its PESQ (1.62) falls *below* the noisy baseline (1.92), indicating active signal degradation. Model D retains 86.6% of its 1-second performance at 25 seconds.

Table 3.6: ESTOI scores across input durations.

Model	1 s	5 s	10 s	15 s	25 s
Noisy	0.79	0.78	0.78	0.77	0.77
Model A	0.91	0.84	0.78	0.74	0.68
Model B	0.90	0.88	0.86	0.85	0.83
Model C	0.92	0.85	0.80	0.75	0.69
Model D	0.92	0.89	0.88	0.86	0.84

Table 3.7: CSIG scores across input durations.

Model	1 s	5 s	10 s	15 s	25 s
Noisy	3.35	3.33	3.32	3.31	3.30
Model A	3.97	3.52	3.18	2.98	2.71
Model B	3.95	3.82	3.74	3.67	3.55
Model C	4.06	3.61	3.27	3.06	2.79
Model D	4.04	3.90	3.82	3.75	3.65

Table 3.8: CBAK scores across input durations.

Model	1 s	5 s	10 s	15 s	25 s
Noisy	2.44	2.43	2.42	2.42	2.41
Model A	3.38	3.02	2.76	2.59	2.38
Model B	3.36	3.24	3.17	3.11	3.02
Model C	3.45	3.10	2.84	2.67	2.46
Model D	3.43	3.32	3.25	3.19	3.12

Table 3.9: COVL scores across input durations.

Model	1 s	5 s	10 s	15 s	25 s
Noisy	2.63	2.61	2.60	2.59	2.58
Model A	3.36	2.90	2.56	2.37	2.11
Model B	3.34	3.21	3.12	3.05	2.93
Model C	3.44	2.99	2.64	2.44	2.18
Model D	3.42	3.29	3.21	3.14	3.04

3.7.2 Performance Degradation Rate Analysis

The relative performance retention (RPR) is defined as:

$$\text{RPR}(d) = \frac{M(d)}{M(1s)} \times 100\%$$

Table 3.10: Relative Performance Retention (%) for PESQ and ESTOI.

Model	PESQ RPR (%)				ESTOI RPR (%)			
	5 s	10 s	15 s	25 s	5 s	10 s	15 s	25 s
Model A	84.3	72.7	66.1	56.6	92.3	85.7	81.3	74.7
Model B	95.4	92.3	89.4	84.9	97.8	95.6	94.4	92.2
Model C	85.0	73.4	66.6	57.3	92.4	87.0	81.5	75.0
Model D	95.5	92.8	90.4	86.6	96.7	95.7	93.5	91.3

Model A retains only 56.6% of its PESQ at 25 seconds; Model D retains 86.6%. The difference between Models A and C is minimal (56.6% vs. 57.3%), confirming that the mask alone does not address length generalization. The CRPE accounts for the dominant improvement (56.6% $\rightarrow$ 84.9%).

3.7.3 Per-SNR Analysis

Table 3.11: PESQ scores at 1-second duration across SNR conditions.

Model	-5 dB	0 dB	5 dB	10 dB	15 dB	20 dB
Noisy	1.26	1.58	1.88	2.18	2.49	2.79
Model A	1.97	2.38	2.76	3.08	3.33	3.62
Model B	1.95	2.36	2.74	3.06	3.31	3.60
Model C	2.06	2.47	2.84	3.14	3.38	3.67
Model D	2.04	2.44	2.82	3.12	3.36	3.65

Table 3.12: PESQ scores at 25-second duration across SNR conditions.

Model	-5 dB	0 dB	5 dB	10 dB	15 dB	20 dB
Noisy	1.24	1.55	1.84	2.13	2.43	2.72
Model A	1.07	1.28	1.51	1.73	1.94	2.19
Model B	1.72	2.06	2.34	2.58	2.79	2.98
Model C	1.12	1.34	1.58	1.80	2.01	2.26
Model D	1.84	2.17	2.45	2.69	2.89	3.07

At 25 seconds, Model A produces PESQ *below* the noisy baseline at all SNR levels from -5 to 10 dB. Models B and D consistently outperform the noisy baseline across all conditions.

3.8 Ablation Study

3.8.1 Effect of Transformer Layers

Table 3.14: Effect of encoder depth on performance at 25 seconds (Model D).

Layers	Params	PESQ	ESTOI	CSIG	CBAK	COVL
4	2.98M	2.36	0.81	3.48	3.01	2.88
6	4.26M	2.52	0.84	3.65	3.12	3.04
8	5.54M	2.55	0.85	3.68	3.14	3.07

3.8.2 Effect of Bias Function

Table 3.15: Effect of bias function on PESQ across durations (Model D).

Bias Function	1 s	5 s	10 s	15 s	25 s
No bias	2.88	2.68	2.53	2.38	2.14
Linear bias	2.87	2.70	2.58	2.44	2.23
Logarithmic (proposed)	2.91	2.78	2.70	2.63	2.52

The logarithmic bias outperforms both alternatives at all durations. The linear bias reaches values $25\times$ larger than training at 25 seconds; the logarithmic bias grows only $1.78\times$.

3.8.3 Effect of Convolution Kernel Size

Table 3.16: Effect of CRPE kernel size on PESQ (Model D).

Kernel Size	1 s	5 s	10 s	15 s	25 s
$k = 3$	2.89	2.74	2.63	2.54	2.40
$k = 7$	2.90	2.76	2.66	2.58	2.46
$k = 15$ (proposed)	2.91	2.78	2.70	2.63	2.52
$k = 31$	2.90	2.77	2.69	2.62	2.50

3.8.4 Effect of Mask Compression Factor

Table 3.17: Effect of SCCM compression factor (Model D).

K	PESQ (1 s)	PESQ (25 s)	ESTOI (1 s)	ESTOI (25 s)
1.0	2.84	2.41	0.90	0.83
1.5	2.88	2.47	0.91	0.83
2.0 (proposed)	2.91	2.52	0.92	0.84
3.0	2.87	2.48	0.91	0.83

Chapter 4. Discussion

This chapter presents a critical analysis of the experimental results reported in Chapter 3.

4.1 Why Relative Positional Embedding Improves Length Generalization

4.1.1 Absolute Positional Encoding and Distribution Shift

The sinusoidal absolute positional encoding creates a distribution shift between training and inference. During training on 1-second segments, the model observes position indices $[0, 62]$. At 25 seconds, indices extend to $[0, 1562]$. The inner products $\langle \text{PE}(t_i), \text{PE}(t_j) \rangle$ at these unseen positions may take values outside the training distribution, causing the softmax to concentrate on unexpected positions.

4.1.2 Relative Representation and Translation Equivariance

The CRPE addresses this through two mechanisms. First, the depthwise separable convolution generates translation-equivariant features—a frame at position 500 receives the same local context as a frame at position 30. Second, the logarithmic bias encodes relative distance rather than absolute position, ensuring consistent attention modulation regardless of absolute index.

4.1.3 Logarithmic Compression and Extrapolation Safety

The ablation study demonstrated that the logarithmic bias substantially outperforms both no-bias and linear-bias alternatives. The linear bias $-\gamma|i - j|$ reaches a maximum of $\gamma \cdot 1562$ at 25 seconds—25× larger than during training. The logarithmic bias reaches only $-\gamma \log(1563) \approx -7.35\gamma$, compared to $-\gamma \log(63) \approx -4.14\gamma$ at training. The extrapolation ratio is only 1.78× versus 25.2×.

4.2 Synergy Between Positional Embedding and Complex Mask

Model D consistently outperforms both Model B and Model C at all durations beyond 1 second. The CRPE provides stable attention patterns that enable the SCCM to exploit its enhanced reconstruction capability reliably. When absolute positional encoding causes erratic attention at long durations (Model C), the complex mask's additional capacity becomes a liability, as the model generates incorrect phase corrections that further distort the signal.

4.3 Analysis of Residual Performance Degradation

Despite substantial improvements, Model D still shows some degradation (PESQ: $2.91 \rightarrow 2.52$, a 13.4% decline). Contributing factors include:

Attention pattern diffusion. The softmax distributes attention over more positions at longer durations, diluting effective weights on the most relevant context.

Statistical distribution shift in content. Training segments contain limited phonetic diversity (single words or phrases). Test signals at 25 seconds contain multiple sentences with diverse prosody.

Noise variability. Longer signals exhibit greater temporal variability in noise characteristics, particularly for non-stationary types like babble and traffic.

4.4 Computational Complexity Analysis

Table 4.1: Computational requirements for different input durations.

Duration	Frames	Attention Size	Memory	Inference (ms)
1 s	63	3,969	< 1 MB	12
5 s	313	97,969	~4 MB	28
10 s	626	391,876	~15 MB	67
15 s	938	879,844	~34 MB	134
25 s	1,563	2,443,969	~93 MB	342

At 25 seconds, inference (342 ms) remains well below signal duration, indicating real-time feasibility. The CRPE adds negligible overhead: $O(T \cdot k \cdot d_{\text{model}})$ is dominated by the $O(T^2 \cdot d_{\text{model}})$ attention computation.

4.5 Comparison with Related Work

At the 1-second in-distribution duration, Model D achieves PESQ = 2.91, competitive with published results including SpeechFormer (2.94), TSTNN (2.96), and SE-Conformer (3.13). The focus of this work is not maximizing in-distribution performance but achieving robust length generalization. To our knowledge, no previously published study has systematically evaluated speech enhancement models across durations from 1 to 25 seconds with models trained exclusively on 1-second segments.

4.6 Metric-Specific Observations

Table 4.2: Relative performance retention of Model D at 25 seconds.

Metric	1 s	25 s	Decline	Retention (%)
PESQ	2.91	2.52	0.39	86.6
ESTOI	0.92	0.84	0.08	91.3
CSIG	4.04	3.65	0.39	90.3
CBAK	3.43	3.12	0.31	91.0
COVL	3.42	3.04	0.38	88.9

PESQ shows the greatest decline (13.4%), while ESTOI and CBAK show the least (~9%). This suggests that perceptual quality is more sensitive to subtle length-extrapolation artifacts than intelligibility or noise suppression, which depend more on local spectral characteristics.

4.7 Limitations

Quadratic complexity. The $O(T^2)$ self-attention prevents application to very long signals (minutes or hours) without efficient attention variants.

No streaming capability. The model requires the entire input before producing output, making it unsuitable for real-time applications.

Fixed STFT resolution. A single time-frequency resolution may not be optimal across all frequency bands.

Single dataset evaluation. Results are limited to VoiceBank-DEMAND with 2 test speakers.

No subjective evaluation. Objective metrics are imperfect proxies for perceived quality.

4.8 Unexpected Findings

Two observations merit particular discussion. First, Model C achieves the highest 1-second performance (PESQ = 2.93) but shows degradation nearly identical to Model A at longer durations. The SCCM and length generalization address orthogonal problems.

Second, Model A's convergence below the noisy baseline was more severe than anticipated. At 25 seconds and -5 dB, Model A's PESQ (1.07) falls substantially below the noisy baseline (1.24), indicating active signal degradation. Analysis

reveals that Model A's attention weights beyond approximately $T = 100$ frames become nearly uniform, effectively averaging features across all positions rather than attending to relevant context.

Chapter 5. Conclusion and Future Work

5.1 Summary of the Research

This thesis investigated the problem of length generalization in Transformer-based speech enhancement. The central question was whether a model trained exclusively on 1-second segments could maintain enhancement performance on substantially longer signals. The investigation was motivated by the observation that conventional Transformer architectures with sinusoidal absolute positional encoding exhibit severe degradation when processing sequences longer than those seen during training—degradation that can render the enhanced output worse than the original noisy signal.

5.2 Summary of the Proposed Method

Two complementary methods were proposed:

1. **Convolutional Relative Positional Embedding (CRPE) with Logarithmic Bias.** Combines depthwise separable convolution (translation-equivariant local features) with a logarithmic distance bias $b_{ij} = \alpha_h \cdot \text{sign}(i - j) \cdot \log(1 + |i - j|)$ that produces bounded attention biases even for distances far exceeding training.
2. **Sigmoid-Compressed Complex Mask (SCCM).** Extends the conventional magnitude-only mask with a compressed magnitude component (bounded by $K = 2$) and a tanh-bounded phase correction term, enabling joint magnitude and phase estimation with training stability.

5.3 Summary of Experimental Findings

Model D (the complete proposed system) achieved PESQ = 2.52 at 25 seconds, retaining 86.6% of its 1-second performance (2.91). The baseline achieved only PESQ = 1.62 at 25 seconds (56.6% retention), falling below the noisy baseline of 1.92. The CRPE with logarithmic bias was the dominant contributor, improving RPR from 56.6% to 84.9%—an improvement of 28.3 percentage points. The SCCM contributed an additional 1.7 percentage points.

5.4 Answers to Research Questions

RQ1: Yes. The proposed CRPE improved PESQ RPR at 25 seconds from 56.6% to 84.9%, consistent across all five metrics and all SNR conditions.

RQ2: The proposed embedding enables stable performance: PESQ declines gradually from 2.84 to 2.41 (Model B), versus a precipitous decline from 2.86 to 1.62 (baseline).

RQ3: Yes. The SCCM improves in-distribution performance (PESQ 2.93 vs. 2.86 at 1 second) through joint magnitude-phase estimation. However, the mask alone does not address length generalization.

RQ4: The combination (Model D) achieves the best overall performance at all durations beyond 1 second through synergistic interaction: the CRPE provides stable attention patterns that enable the SCCM to exploit its reconstruction capability reliably.

RQ5: Model D retains approximately 95.5% of PESQ at 5 seconds, 92.8% at 10 seconds, 90.4% at 15 seconds, and 86.6% at 25 seconds.

RQ6: The CRPE contributes 28.3 percentage points of RPR improvement; the SCCM contributes 1.7 additional points. The components address orthogonal aspects of the problem.

5.5 Contributions of This Thesis

1. A Transformer-based speech enhancement architecture optimized for mask-based enhancement in the STFT domain.
2. A convolutional relative positional embedding with logarithmic distance bias that significantly improves length generalization.
3. A sigmoid-compressed complex mask that jointly estimates magnitude and phase corrections with bounded stability.
4. A systematic experimental framework for evaluating length generalization by training on 1-second segments and testing on durations up to 25 seconds.
5. A comprehensive ablation study demonstrating the dominant contribution of positional embedding (28.3 pp) over mask design (1.7 pp) to length generalization.
6. A quantitative evaluation using five complementary metrics across six SNR conditions and five input durations.
7. An empirical demonstration that absolute positional encoding causes catastrophic failure at unseen lengths, while the proposed relative embedding avoids this failure mode entirely.

5.6 Future Work

Variable-length training with curriculum learning. Progressive exposure to longer sequences during training could reduce residual degradation at long durations.

Efficient attention mechanisms. Linear attention, sparse attention, or chunked attention could extend applicability to signals of minutes or hours.

Streaming speech enhancement. Causal attention with block-wise processing would enable real-time applications with bounded latency.

Phase-aware complex-valued networks. Complex-valued layers maintaining algebraic structure throughout the network could improve phase modeling beyond the decoupled SCCM approach.

Larger and more diverse datasets. Cross-corpus evaluation on DNS Challenge, LibriSpeech + noise augmentation, and WHAM! would strengthen generalizability claims.

Rotary positional embedding for audio. Adapting RoPE with frequency-domain rotary encodings that respect spectral structure is a natural extension.

Knowledge distillation. Compact student models retaining length-generalization properties would enable deployment on resource-constrained devices.

Cross-domain generalization. Investigating whether improved temporal modeling also benefits generalization to unseen noise types and recording conditions.

5.7 Concluding Remarks

This thesis has demonstrated that the choice of positional representation in Transformer-based speech enhancement has a profound impact on the model's ability to process input sequences of varying length. The conventional sinusoidal absolute positional encoding creates a hard coupling between the model's attention mechanism and the training sequence lengths, leading to catastrophic failure at unseen lengths. The proposed convolutional relative positional embedding with logarithmic bias breaks this coupling by representing position in terms of local context and relative distance rather than absolute index, enabling robust generalization from 1-second training segments to inputs up to 25 seconds.

The practical implication is that Transformer-based speech enhancement systems can be designed to handle variable-length inputs gracefully, without requiring fixed-length chunking or retraining on longer segments. By appropriately encoding positional information, a model trained on readily available short segments can be deployed on substantially longer signals with only modest performance degradation. As Transformer architectures continue to advance in speech processing, the principles demonstrated here—relative positional representation, bounded extrapolation, and translation-equivariant local context—will remain relevant for designing models that generalize robustly beyond their training conditions.

References

1. Ba, J. L., Kiros, J. R., and Hinton, G. E. (2016). Layer normalization. *arXiv preprint arXiv:1607.06450*.
2. Berouti, M., Schwartz, R., and Makhoul, J. (1979). Enhancement of speech corrupted by acoustic noise. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 208–211.
3. Boll, S. (1979). Suppression of acoustic noise in speech using spectral subtraction. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 27(2), pp. 113–120.
4. Chen, S., Wu, Y., Chen, Z., Wu, J., Li, J., Yoshioka, T., Wang, C., Liu, S., and Zhou, M. (2022). Continuous speech separation with conformer. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 6552–6556.
5. Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., and Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1724–1734.
6. Choi, H., Kim, J., Huh, J., Kim, A., Ha, J., and Lee, K. (2019). Phase-aware speech enhancement with deep complex U-Net. *Proceedings of the International Conference on Learning Representations (ICLR)*.
7. Dai, Z., Yang, Z., Yang, Y., Carbonell, J., Le, Q. V., and Salakhutdinov, R. (2019). Transformer-XL: Attentive language models beyond a fixed-length context. *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 2978–2988.
8. Ephraim, Y. and Malah, D. (1984). Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 32(6), pp. 1109–1121.
9. Ephraim, Y. and Malah, D. (1985). Speech enhancement using a minimum mean-square error log-spectral amplitude estimator. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 33(2), pp. 443–445.
10. Ephraim, Y. and Van Trees, H. L. (1995). A signal subspace approach for speech enhancement. *IEEE Transactions on Speech and Audio Processing*, 3(4), pp. 251–266.
11. Erdogan, H., Hershey, J. R., Watanabe, S., and Le Roux, J. (2015). Phase-sensitive and recognition-boosted speech separation using deep recurrent neural networks.

- Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 708–712.
12. Fu, S., Hu, T., Tsao, Y., and Lu, X. (2018). Complex spectrogram enhancement by convolutional neural network with multi-metrics learning. *Proceedings of the IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, pp. 1–6.
 13. Gulati, A., Qin, J., Chiu, C., Parmar, N., Zhang, Y., Yu, J., Han, W., Wang, S., Zhang, Z., Wu, Y., and Pang, R. (2020). Conformer: Convolution-augmented Transformer for speech recognition. *Proceedings of Interspeech*, pp. 5036–5040.
 14. He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778.
 15. Hendrycks, D. and Gimpel, K. (2016). Gaussian error linear units (GELUs). *arXiv preprint arXiv:1606.08415*.
 16. Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), pp. 1735–1780.
 17. Hu, Y. and Loizou, P. C. (2008). Evaluation of objective quality measures for speech enhancement. *IEEE Transactions on Audio, Speech, and Language Processing*, 16(1), pp. 229–238.
 18. Hu, Y., Liu, Y., Lv, S., Xing, M., Zhang, S., Fu, Y., Wu, J., Zhang, B., and Xie, L. (2020). DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement. *Proceedings of Interspeech*, pp. 2472–2476.
 19. Jansson, A., Humphrey, E. J., Montecchio, N., Bittner, R., Kumar, A., and Weyde, T. (2017). Singing voice separation with deep U-Net convolutional networks. *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, pp. 745–751.
 20. Jensen, J. and Taal, C. H. (2016). An algorithm for predicting the intelligibility of speech masked by modulated noise maskers. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 24(11), pp. 2009–2022.
 21. Khandelwal, U., He, H., Qi, P., and Jurafsky, D. (2018). Sharp nearby, fuzzy far away: How neural language models use context. *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 284–294.
 22. Kazemnejad, A., Padhi, I., Natesan Ramamurthy, K., Das, P., and Reddy, S. (2024). The impact of positional encoding on length generalization in Transformers. *Advances in Neural Information Processing Systems (NeurIPS)*, 36.
 23. Kim, J., El-Khamy, M., and Lee, J. (2020). T-GSA: Transformer with Gaussian-weighted self-attention for speech enhancement. *Proceedings of the IEEE*

- International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 6649–6653.
24. Koizumi, Y., Yatabe, K., Delcroix, M., Masuyama, Y., and Takeuchi, D. (2020). Speech enhancement using self-adaptation and multi-head self-attention. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 181–185.
 25. Le Roux, J., Wisdom, S., Erdogan, H., and Hershey, J. R. (2019). SDR—half-baked or well done? *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 626–630.
 26. Lim, J. S. and Oppenheim, A. V. (1979). Enhancement and bandwidth compression of noisy speech. *Proceedings of the IEEE*, 67(12), pp. 1586–1604.
 27. Loizou, P. C. (2007). *Speech Enhancement: Theory and Practice*. CRC Press, Boca Raton, FL.
 28. Luo, Y. and Mesgarani, N. (2019). Conv-TasNet: Surpassing ideal time-frequency magnitude masking for speech separation. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 27(8), pp. 1256–1266.
 29. Meyes, R., Lu, M., de Puiseau, C. W., and Meisen, T. (2019). Ablation studies in artificial neural networks. *arXiv preprint arXiv:1901.08644*.
 30. Narayanan, A. and Wang, D. (2013). Ideal ratio mask estimation using deep neural networks for robust speech recognition. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 7092–7096.
 31. Paliwal, K., Wojcicki, K., and Shannon, B. (2011). The importance of phase in speech enhancement. *Speech Communication*, 53(4), pp. 465–494.
 32. Park, S. R. and Lee, J. (2017). A fully convolutional neural network for speech enhancement. *Proceedings of Interspeech*, pp. 1993–1997.
 33. Press, O., Smith, N. A., and Lewis, M. (2022). Train short, test long: Attention with linear biases enables input length extrapolation. *Proceedings of the International Conference on Learning Representations (ICLR)*.
 34. Rix, A. W., Beerends, J. G., Hollier, M. P., and Hekstra, A. P. (2001). Perceptual evaluation of speech quality (PESQ)—A new method for speech quality assessment of telephone networks and codecs. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 749–752.
 35. Ronneberger, O., Fischer, P., and Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 234–241.

36. Shaw, P., Uszkoreit, J., and Vaswani, A. (2018). Self-attention with relative position representations. *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pp. 464–468.
37. Srinivasan, S., Roman, N., and Wang, D. (2006). Binary and ratio time-frequency masks for robust speech recognition. *Speech Communication*, 48(11), pp. 1486–1501.
38. Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., and Liu, Y. (2024). RoFormer: Enhanced Transformer with rotary position embedding. *Neurocomputing*, 568, p. 127063.
39. Subakan, C., Ravanelli, M., Cornell, S., Bronzi, M., and Zhong, J. (2021). Attention is all you need in speech separation. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 21–25.
40. Tan, K. and Wang, D. (2019). A convolutional recurrent neural network for real-time speech enhancement. *Proceedings of Interspeech*, pp. 3229–3233.
41. Thiemann, J., Ito, N., and Vincent, E. (2013). The diverse environments multi-channel acoustic noise database (DEMAND): A database of multichannel environmental noise recordings. *Journal of the Acoustical Society of America*, 133(5), p. 3591.
42. Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A., Grave, E., and Lample, G. (2023). LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.
43. Valentini-Botinhao, C., Wang, X., Takaki, S., and Yamagishi, J. (2016). Investigating RNN-based speech enhancement methods for noise-robust text-to-speech. *Proceedings of the ISCA Speech Synthesis Workshop (SSW)*, pp. 146–152.
44. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, pp. 5998–6008.
45. Veaux, C., Yamagishi, J., and King, S. (2013). The voice bank corpus: Design, collection and data analysis of a large regional accent speech database. *Proceedings of O-COCOSDA/CASLRE*, pp. 1–4.
46. Wang, D. (2005). On ideal binary mask as the computational goal of auditory scene analysis. In *Speech Separation by Humans and Machines*, Springer, pp. 181–197.
47. Wang, Y. and Wang, D. (2012). Towards scaling up classification-based speech separation. *IEEE Transactions on Audio, Speech, and Language Processing*, 21(7), pp. 1381–1390.
48. Wang, Y., Narayanan, A., and Wang, D. (2014). On training targets for supervised speech separation. *IEEE/ACM Transactions on Audio, Speech, and Language*

- Processing*, 22(12), pp. 1849–1858.
49. Wang, Z., Wang, S., Li, Z., Li, Z., Li, Z., and Zheng, H. (2021). TSTNN: Two-stage Transformer based neural network for speech enhancement in the time domain. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 7098–7102.
 50. Weninger, F., Erdogan, H., Watanabe, S., Vincent, E., Le Roux, J., Hershey, J. R., and Schuller, B. (2015). Speech enhancement with LSTM recurrent neural networks and its application to noise-robust ASR. *Proceedings of the International Conference on Latent Variable Analysis and Signal Separation (LVA/ICA)*, pp. 91–99.
 51. Wiener, N. (1949). *Extrapolation, Interpolation, and Smoothing of Stationary Time Series*. MIT Press, Cambridge, MA.
 52. Wilson, K. W., Raj, B., Smaragdis, P., and Divakaran, A. (2008). Speech denoising using nonnegative matrix factorization with priors. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 4029–4032.
 53. Williamson, D. S., Wang, Y., and Wang, D. (2016). Complex ratio masking for monaural speech separation. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 24(3), pp. 483–492.
 54. Xiong, R., Yang, Y., He, J., Zheng, K., Zheng, S., Xing, C., Zhang, H., Lan, Y., Wang, L., and Liu, T. (2020). On layer normalization in the Transformer architecture. *Proceedings of the International Conference on Machine Learning (ICML)*, pp. 10524–10533.
 55. Xu, Y., Du, J., Dai, L., and Lee, C. (2015). A regression approach to speech enhancement based on deep neural networks. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 23(1), pp. 7–19.
 56. Yin, D., Luo, C., Xiong, Z., and Zeng, W. (2020). PHASEN: A phase-and-harmonics-aware speech enhancement network. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05), pp. 9458–9465.